

Hybrid Quantum Traffic Optimization Framework Using Digital Twins for Major Public Events.

PHASE I REPORT

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI

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October 2026

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BONAFIDE CERTIFICATE

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“This project addresses the following Sustainable Development Goal(s): **Goal- No :11:aims to make cities and human settlements inclusive, safe, resilient, and sustainable** ”

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ABSTRACT

Major public events such as sports tournaments, concerts, festivals, and religious gatherings often generate sudden increases in vehicular and pedestrian traffic, placing significant pressure on urban transportation networks. These traffic surges can lead to severe congestion, inefficient traffic signal coordination, increased travel delays, and challenges for emergency vehicle movement. Although Digital Twin technology has emerged as an effective solution for monitoring and simulating urban traffic, most existing frameworks are primarily designed for routine traffic conditions and provide limited support for event-specific traffic management. Additionally, conventional optimization techniques face challenges in handling the large-scale, dynamic, and multi-objective nature of traffic scenarios associated with major public events. To address these limitations, this paper proposes a Hybrid Quantum Traffic Optimization Framework Using Digital Twins for Major Public Events. The proposed framework integrates a Digital Twin-based traffic simulation environment with hybrid quantum-classical optimization to model, analyze, and optimize traffic conditions before and during planned events. The Digital Twin creates a virtual representation of the transportation network by incorporating road infrastructure, traffic signals, event schedules, and mobility patterns of both vehicles and pedestrians. The optimization component formulates traffic management as a constrained optimization problem using a Quadratic Unconstrained Binary Optimization (QUBO) model, enabling efficient exploration of traffic signal timing, route allocation, and congestion management strategies through hybrid quantum computing. The proposed framework aims to provide traffic authorities with a proactive decision-support system for evaluating multiple traffic management strategies within a realistic virtual environment before real-world deployment.

ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S.MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.**, and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.**, our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. J. Manoranjini, Ph.D.**, Professor and Head of the Department of Computer Science and Engineering for his guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide, **Dr.V.Karthick**, Associate Professor, Department of Computer Science and Engineering for his valuable guidance throughout the course of the project. We thank our Project Coordinator, **Dr.S.Madhusudhanan** Assistant Professor, Department of Computer Science and Engineering for his useful tips during our review to build our project.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
DGMARL	Decentralized Graph-based Multi-Agent Reinforcement Learning
D-Wave	Quantum Computing System / Quantum Annealing Platform
FedQ-Twin	Federated Quantum Digital Twin
GNN-LSTM	Graph Neural Network – Long Short-Term Memory
GPS	Global Positioning System
IPL	Indian Premier League
MLK	Martin Luther King
OSMnx	OpenStreetMap NetworkX
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analysis
QAOA	Quantum Approximate Optimization Algorithm
QUBO	Quadratic Unconstrained Binary Optimization
SUMO	Simulation of Urban Mobility

CHAPTER 1

INTRODUCTION

1.1 GENERAL

The rapid growth of urban populations and increasing vehicle ownership have created significant challenges in managing traffic flow, particularly in densely populated cities. These challenges become more prominent during large public events such as sports matches, concerts, festivals, and religious gatherings, where a large number of vehicles are expected to reach a common destination within a limited period. The resulting concentration of traffic can lead to congestion, increased travel time, fuel consumption, and delays. Therefore, effective event-based traffic management requires advance planning of routes, intersections, signal timings, and traffic restrictions.

Traditional traffic management approaches generally depend on predefined regulations, fixed signal timings, manual planning, and experience-based decisions. However, event-based traffic management involves several interconnected decisions, including route diversions, signal timing adjustments, and temporary traffic restrictions, while considering expected vehicle demand and available road capacity. The combination of these decisions forms a complex optimization problem.

The Quantum Traffic Optimizer is proposed as a pre-event traffic planning framework that applies quantum optimization to anticipated traffic conditions associated with a large public event. The framework considers the surrounding road network, expected vehicle demand, traffic corridors, signal timings, route-diversion possibilities, and temporary traffic restrictions to generate a suitable traffic-management plan before the event. The system is intended as a decision-support tool for traffic authorities and integrates traffic-network modelling, event-based demand estimation, behavioural modelling, mathematical optimization, and quantum optimization.

1.2 PROBLEM STATEMENT

Managing traffic during large public events is a challenging task due to the sudden increase in vehicle demand within a limited geographical area and time period. Conventional traffic planning methods often rely on fixed signal timings, predefined routes, manual decisions, and isolated traffic-control measures, which may not adequately address the complex interactions

between multiple roads, intersections, and vehicle movements. Before an event, traffic authorities need to determine how the expected vehicle demand should be distributed across available routes and how traffic signals and temporary restrictions should be adjusted to reduce congestion. Since several possible traffic-management actions can be selected simultaneously, identifying an effective combination of decisions becomes a complex combinatorial optimization problem.

The absence of a systematic pre-event optimization approach can make it difficult to evaluate different traffic-management strategies and select suitable decisions before congestion occurs. Therefore, there is a need for a framework that can model the expected traffic conditions of a planned event and optimize multiple traffic-management decisions while considering the characteristics and constraints of the surrounding road network. The Quantum Traffic Optimizer addresses this problem by formulating route diversions, signal timing adjustments, and temporary traffic restrictions as binary decision variables and representing their relationships through a QUBO-based optimization model. Quantum optimization is then applied to identify a suitable combination of decisions that can be interpreted as a pre-event traffic-management plan for the selected event scenario.

1.3 OBJECTIVE

The primary objective of the Quantum Traffic Optimizer is to develop a pre-event traffic planning framework that uses quantum optimization to identify suitable traffic-management decisions for large public events. The system aims to model anticipated vehicle demand and road-network conditions, formulate traffic-management actions as an optimization problem, and generate a practical plan involving route diversions, signal timing adjustments, and temporary traffic restrictions.

The specific objectives are:

1. To model the road network surrounding a selected large public-event location using real-world road and intersection data.
2. To estimate the expected vehicle demand and arrival patterns associated with the planned event.
3. To identify relevant traffic-management decisions, including route diversions, signal extensions, and temporary traffic restrictions.
4. To formulate the traffic-management problem using Quadratic Unconstrained Binary Optimization (QUBO).
5. To apply quantum optimization techniques, particularly QAOA, to obtain an optimized combination of traffic-management decisions.

6. To decode the optimized solution into practical traffic-management actions that can be implemented before the event.
7. To compare the optimized solution with a classical optimization approach based on objective value, traffic measures, and computational performance.
8. To evaluate the potential of quantum optimization as a decision-support approach for pre-event traffic planning.

1.4 EXISTING SYSTEM

An existing approach for event-based traffic optimization is the Quantum Shuttle framework proposed by Yarkoni et al. (2020). The framework addresses vehicle route assignment during large-scale events by generating candidate routes and formulating the route-selection problem as a Quadratic Unconstrained Binary Optimization (QUBO) problem. Quantum optimization is then used to obtain a suitable combination of vehicle routes while reducing overlap between their paths.

The framework combines event-related transportation data, vehicle information, candidate-route generation, and QUBO-based optimization. It demonstrates the application of quantum computing to transportation problems associated with large events.

However, the Quantum Shuttle framework primarily focuses on vehicle route assignment and does not directly address a broader pre-event traffic-management plan involving multiple decisions such as route diversions, signal timing adjustments, and temporary traffic restrictions. The study also reported limitations related to quantum-hardware scalability, qubit connectivity, and noise. These aspects indicate the need for a framework that considers multiple traffic-management decisions together during pre-event planning.

The Quantum Shuttle framework formulates vehicle route selection as a QUBO problem using binary decision variables. Let $q_{i,j}$ represent the route-selection decision, where i denotes a vehicle and j denotes one of its candidate routes. The variable is 1 when route j is selected for vehicle i and 0 otherwise. .

The objective function is formulated as:

$$Obj = \sum_{p \in P} cost(p) + \lambda \sum_i \left(\sum_j q_{i,j} - 1 \right)^2$$

where P represents the set of shared GPS points among the candidate route. The first term represents the route-overlap cost, while the second term is a penalty term that ensures that exactly one route is selected for each vehicle. The parameter λ controls the strength of this constraint.

The cost associated with a shared point p is calculated as:

$$cost(p) = \left(\sum_{q_{ij} \in B(p)} q_{ij} \right)^2$$

where $B(p)$ represents the set of route-selection variables corresponding to routes passing through GPS point p . Therefore, routes with greater overlap contribute a larger cost to the objective function.

1.5 PROPOSED SYSTEM

The proposed Quantum Traffic Optimizer is a pre-event traffic planning system designed to assist authorities in preparing for the expected traffic conditions associated with large public events. The system focuses on situations where a significant number of vehicles are expected to travel towards an event location within a limited period. Instead of responding to congestion only after it occurs, the proposed system aims to support advance planning by analysing the anticipated traffic conditions and identifying suitable traffic-management decisions for the event.

The system considers the surrounding road network and the expected movement of vehicles associated with the event to evaluate possible traffic-management alternatives. These alternatives may include changes to vehicle routes, adjustments to traffic signal timings, and temporary restrictions on selected roads or corridors. The system is designed to consider these decisions together rather than treating each traffic-control measure independently, allowing different combinations of possible actions to be evaluated according to the requirements of the selected event scenario.

The Quantum Traffic Optimizer provides the resulting traffic-management plan as a decision-support output for traffic authorities. The generated plan is intended to help authorities understand which traffic-management actions can be considered before the event and how these actions may affect the anticipated traffic conditions. The proposed system therefore provides a structured approach for pre-event traffic planning and demonstrates the potential use of quantum computing techniques in addressing complex urban traffic-management problems.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

Yarkoni et al. proposed a quantum-based approach for managing bus routes during the Web Summit 2019 in Lisbon. Candidate routes were generated using real-time bus-location and traffic information, and route selection was formulated as a QUBO problem using binary variables. A D-Wave 2000Q processor was used with classical computing components for data preparation and route generation. The system was tested with nine buses during the event and demonstrated the feasibility of quantum-assisted route planning. However, the small-scale deployment and limitations of the available quantum hardware restricted its application to larger transportation networks.[3]

Kumarasamy et al. developed a Digital Twin and reinforcement-learning approach for traffic-signal control at 11 intersections in the MLK Smart Corridor, Chattanooga. A PTV-Vissim Digital Twin was used to reproduce traffic conditions and evaluate signal-control decisions, while learning agents exchanged information with neighbouring intersections. The approach reduced traffic-related delays and stops compared with conventional actuated signal control. However, information-transfer delays between the physical and digital systems and the need for evaluation on larger networks were identified as limitations [2].

Mohammed et al. reviewed the use of quantum annealing for transportation problems including traffic-signal control, traffic-flow optimization, and vehicle routing. The reviewed approaches commonly used QUBO and Ising formulations and were implemented on platforms such as D-Wave systems and digital annealers. The studies reported improvements in measures such as waiting time, traffic flow, and travel distance under specific conditions. However, most evaluations were simulation-based, with qubit connectivity, hardware capacity, noise, and problem size remaining important challenges [6].

Faliagka et al. proposed a Digital Twin framework that integrates information from sensors, vehicles, applications, and other sources into a shared digital environment. The framework supports traffic monitoring and what-if analysis before changes are applied to the physical system, with applications including smart parking and emergency management. However, the work mainly focuses on Digital Twin architecture rather than a specific traffic-optimization method. Technology integration, implementation cost, privacy, and infrastructure maturity were also identified as challenges [8].

Dribi et al. conducted a PRISMA-based systematic review of 165 studies on real-time intelligent traffic management in smart cities. The review found that dynamic traffic-control methods were widely adopted, while Digital Twin integration remained limited. Key challenges included scalability, heterogeneous sensor data, interoperability, latency, privacy, and limited real-world

validation. The authors highlighted the need for integrated traffic-management systems combining sensing, AI, optimization, and real-time control [5].

Kumarasamy et al. further proposed a Digital Twin-assisted Decentralized Graph-based Multi-Agent Reinforcement Learning (DGMARL) method for traffic-signal optimization. The approach used learning agents at individual intersections that exchanged traffic-state information, with a PTV-Vissim Digital Twin supporting training and evaluation across an 11-intersection corridor. The method achieved a 44.27% reduction in Eco-PI over 24 hours and 29.88% during PM peak hours compared with actuated signal control. However, it focuses on reinforcement learning and does not incorporate quantum-based optimization or federated learning [7].

Kim et al. investigated quantum annealing for large optimization problems, focusing on limitations related to qubit availability and connectivity. The study compared QUBO-based approaches with classical methods and reported faster solution times for the evaluated large-scale problems using hybrid quantum annealing. For a 10,000-variable problem, the reported execution time for Hybrid Quantum Annealing was 0.0855 seconds, compared with 101 seconds for QA-QBSolv and 561 seconds for another evaluated method. Since the study was not transportation-specific, traffic factors such as vehicle waiting time and pedestrian movement were not considered [4].

Prabaharan proposed FedQ-Twin, which combines federated learning using GNN-LSTM, quantum-inspired optimization, and Digital Twin simulation for traffic management. Federated learning supports collaborative traffic prediction without directly sharing underlying traffic data, while the optimization component uses an Ising formulation for traffic-control decisions. The Digital Twin provides a simulation environment for evaluating these decisions. However, the work is presented as a preprint and requires further evaluation using real-world traffic datasets and physical traffic infrastructure [1].

The reviewed studies show progress in quantum optimization, Digital Twin technology, and AI-based traffic management, but these approaches are generally applied to separate aspects of transportation or specific traffic problems [3], [6], [8]. Quantum-based studies mainly focus on QUBO or Ising formulations, while Digital Twin and reinforcement-learning studies focus on simulation, prediction, and traffic-signal control [2], [6]. Although integrated frameworks such as FedQ-Twin combine multiple technologies, they remain largely simulation-based [1]. Limited work has specifically addressed the use of Digital Twin-based traffic simulation with quantum optimization for selecting traffic-management strategies during major public events [1], [3], [8]. Challenges related to scalability, hardware constraints, latency, and real-world validation have also been reported across the reviewed studies [4]–[6], [8].

Despite the progress reported in existing studies, limited work has investigated the integration of Digital Twin-based traffic simulation with quantum optimization for selecting traffic-management strategies during major public events. Existing studies also highlight challenges related to scalability, quantum hardware constraints, communication latency, and

real-world validation. These gaps motivate the development of a framework that can evaluate alternative traffic strategies within a simulated event environment.

To address this gap, this study proposes a hybrid quantum-classical traffic optimization framework integrated with a Digital Twin environment for major public events. The framework models event-specific traffic conditions and uses quantum-based optimization to evaluate alternative traffic-management strategies before their potential application to the real traffic network.

2.2 COMPARISON AND DISCUSSION

Existing traffic-management approaches for large public events commonly use predefined regulations, fixed or rule-based control strategies, historical information, and manual decisions. Although these approaches can support known traffic conditions, evaluating multiple interconnected decisions such as route diversions, signal timing, and temporary road restrictions can become difficult when event-related traffic demand is high.

The proposed Quantum Traffic Optimizer provides an optimization-based approach for evaluating these decisions together during pre-event planning. The system considers expected traffic conditions and road-network constraints and generates a suitable combination of traffic-management actions. It is intended as a decision-support system that complements existing traffic authorities and operational procedures rather than replacing them.

2.3 CONCLUSION

The literature reviewed in this chapter shows that quantum optimization, Digital Twin technology, and AI-based traffic management have been applied to different transportation problems. However, limited work has focused on integrating Digital Twin-based traffic simulation with quantum optimization for pre-event traffic planning during major public events. The identified challenges related to scalability, quantum hardware, latency, and real-world validation provide the basis for developing the proposed Quantum Traffic Optimizer as a decision-support framework for evaluating traffic-management strategies before an event.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The proposed Hybrid Quantum Traffic Optimization Framework uses a SUMO-based Digital Twin for pre-event traffic planning around M.A. Chidambaram Stadium. The architecture combines microscopic traffic simulation, traffic-state processing, quantum optimization, and 3D visualization. Event information, road-network information, and expected traffic demand are processed before generating traffic-management decisions.

Simulation of Urban MObility (SUMO) is used as the microscopic traffic simulation environment. It represents vehicles, pedestrians, traffic signals, and public transport within the selected road network. The simulation provides the traffic state required for evaluating congestion and testing traffic-management decisions.

The Digital Twin provides a virtual representation of the selected stadium-area traffic environment. In this system, the Digital Twin is simulation-driven, where the traffic state is generated by SUMO and transferred to the visualization layer. This allows different event scenarios to be simulated, optimized, and observed in a controlled environment.

TraCI (Traffic Control Interface) connects SUMO with the Python backend. Through TraCI, the backend retrieves simulation information and applies traffic-control decisions such as route changes and signal-timing modifications. The Python backend also performs traffic-state processing and connects the optimization module with the simulation.

The optimized decisions are mapped back to the SUMO network through TraCI. The updated simulation state is transmitted through WebSocket communication to a Three.js-based Digital Twin, which provides a 3D representation of the traffic environment.

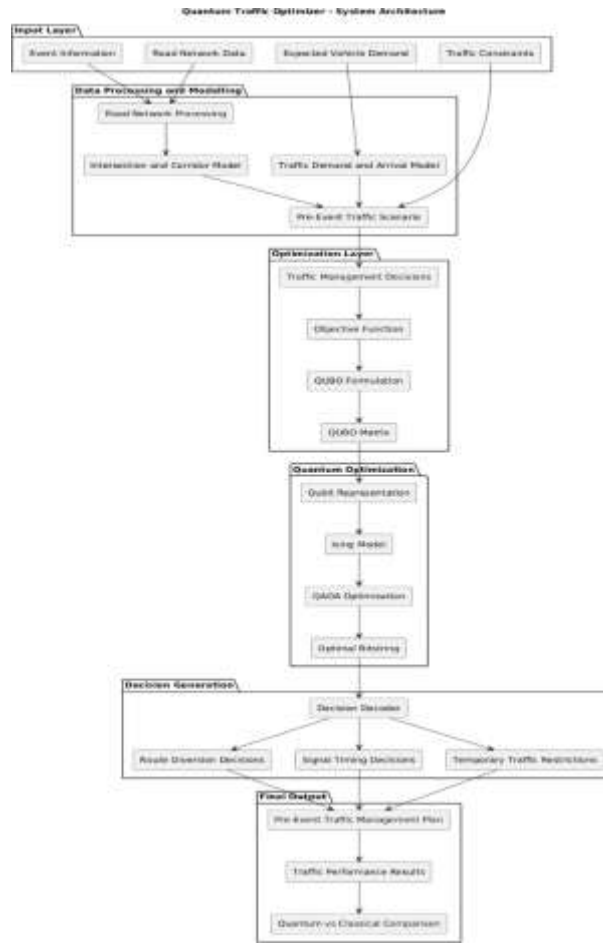


Figure 1 : Overall System Architecture

The architecture can be represented through the following major stages:

1. **Input Layer:** Provides event information, road-network data, expected vehicle demand, and traffic constraints.
2. **SUMO Simulation Layer:** Represents roads, vehicles, pedestrians, traffic signals, and the elevated railway within the simulation environment.
3. **TraCI and Processing Layer:** Enables the Python backend to retrieve simulation states and traffic metrics from SUMO.
4. **Traffic Scenario Modelling:** Represents Normal Day and Event Day conditions and the expected traffic behaviour.

5. Optimization Problem Formulation: Represents route diversions, signal adjustments, and temporary restrictions as binary decisions.
6. Quantum Optimization: Processes the QUBO formulation using the quantum optimization approach.
7. Traffic Control Integration: Maps the selected decisions to the corresponding SUMO roads and junctions through TraCI.
8. Digital Twin Visualization: Sends the updated simulation state through WebSocket to the Three.js visualization layer.
9. Traffic Management Plan: Provides the resulting traffic-management actions for pre-event planning and evaluation.

3.2 SYSTEM REQUIREMENTS

3.2.1 SOFTWARE REQUIREMENTS

The main software components are Python for backend processing and optimization integration, SUMO for microscopic traffic simulation, TraCI for communication between SUMO and Python, OSMnx and NetworkX for road-network processing, NumPy and Pandas for numerical and data processing, Qiskit for quantum circuit and QAOA implementation, FastAPI for backend services, WebSocket for simulation-state communication, and Three.js/WebGL for Digital Twin visualization. VS Code and Jupyter Notebook are used as development environments. The optimization component can use software-based quantum simulators or classical backends for experimentation.

3.2.2 HARDWARE REQUIREMENTS

The Quantum Traffic Optimizer does not require specialized quantum hardware. A standard computer or laptop with a modern multi-core processor, at least 8 GB RAM, and sufficient storage is adequate for the current prototype. Internet connectivity may be required for obtaining road-network data from OpenStreetMap, while subsequent processing and optimization can be performed locally. Software-based quantum simulation or classical backends are used for the current experiments.

3.3 NETWORK AND DATA DESIGN

3.3.1 Event and Traffic Data

The study considers traffic conditions associated with an event at M.A. Chidambaram Stadium, Chepauk. Event information includes the event location, expected vehicle demand, arrival patterns, and traffic-management requirements. Both Normal Day and Event Day conditions are considered. Event-day demand represents increased vehicle and pedestrian

activity around the stadium and associated transport locations. The event information and traffic demand are provided as inputs to the SUMO simulation.

3.3.2 Road Network Data

The road network is obtained from OpenStreetMap (OSM) and processed using OSMnx. The study area contains approximately 750 intersections and 1,985 road segments. The processed network includes roads, junctions, traffic signals, pedestrian movement, and relevant public-transport infrastructure. The network is converted into a SUMO-compatible representation for microscopic traffic simulation and forms the underlying environment of the Digital Twin.

3.3.3 Intersection and Road Representation

The road network is represented as a graph consisting of nodes and edges. Intersections are represented as nodes, while road segments are represented as edges. Each element is associated with corresponding SUMO identifiers. The optimization module uses these identifiers to connect its decisions with the simulation. Route, signal, and traffic-restriction decisions are mapped to the corresponding SUMO elements through TraCI. This mapping also allows the same traffic-control changes to be represented in the Digital Twin.

3.3.4 Traffic Corridor Identification

Traffic corridors represent groups of connected roads that are important for movement around the event location. Four major corridors are considered:

- Anna Salai
- Triplicane High Road
- Wallajah Road
- Kamarajar Salai

These corridors contain a combined 131 mapped road segments. They are used to define route-diversion and temporary-restriction decisions at the corridor level.

3.4 TRAFFIC DEMAND AND BEHAVIOURAL MODEL

The traffic scenario combines expected event demand with microscopic traffic behaviour in SUMO. The Digital Twin represents vehicles, pedestrians, traffic signals, and the elevated MRTS railway under Normal Day and Event Day conditions.

3.4.1 Event Vehicle Demand Estimation

The Quantum Traffic Optimizer requires an estimate of the number of vehicles expected to travel towards the event location during the selected planning period. For the experimental scenario, an IPL match at the M.A. Chidambaram Stadium, Chennai is considered, with an expected total demand of 9,515 vehicles. This demand represents the anticipated event-related vehicle movement and is distributed among selected entry points surrounding the study area. The estimation provides a common demand input for evaluating the traffic conditions that may occur before the event.

The estimated demand is divided among five major entry points, represented as J1, J2, J3, J4, and J5, with respective vehicle demands of 1,200, 1,447, 600, 1,100, and 450 vehicles for the defined planning scenario. These values provide the initial traffic demand associated with different parts of the surrounding network. The demand values are used as planning inputs rather than as real-time measurements, allowing the system to model anticipated event traffic and evaluate possible traffic-management decisions before the event takes place.

3.4.2 Temporal Arrival Distribution

Event-related traffic does not necessarily arrive at the destination at the same time. Vehicles are expected to arrive progressively during the period leading up to the event, resulting in variations in traffic demand across different time intervals. The Quantum Traffic Optimizer therefore represents the expected vehicle arrivals across the planning period rather than assuming that the complete event demand occurs simultaneously. This provides a more realistic representation of the buildup of traffic towards the event location.

The temporal distribution is used to understand how the anticipated demand may affect the road network as the event approaches. By representing vehicle arrivals over time, the system can identify periods in which particular intersections or corridors may experience higher demand. This temporal representation supports the evaluation of traffic-management decisions such as route diversions and signal adjustments under changing anticipated demand conditions.

3.4.3 Vehicle Demand at Intersections

Vehicle demand is associated with selected intersections to represent the locations through which event traffic passes. Intersections J1–J10 are mapped to corresponding SUMO junctions. This representation allows anticipated demand to be related to individual road sections and intersections and provides a basis for signal-related optimization decisions.

3.4.4 Traffic Scenario and Constraint Representation

The framework considers Normal Day and Event Day scenarios. SUMO represents microscopic vehicle movement, pedestrian movement, traffic signals, and the elevated railway. Vehicles are modelled using the Intelligent Driver Model (IDM), while pedestrians use defined movement paths and the railway follows its configured operational behaviour.

For IDM, the main parameters include desired speeds of 13.89 m/s on major roads and 11.11 m/s on minor roads, maximum acceleration of 2.6 m/s², comfortable deceleration of 4.5 m/s², jam distance of 2.0 m, and safe time headway of 1.2 s.

The elevated Chepauk MRTS is represented as a grade-separated railway, avoiding road-level crossings. Trains follow the configured railway model and station dwell behaviour. When trains arrive at Chepauk Elevated Station, additional pedestrian demand can be generated toward the stadium area, allowing the Digital Twin to represent multimodal event conditions.

Signal decisions are constrained using a minimum green time of 30 seconds, maximum green time of 60 seconds, and yellow interval of 5 seconds. These constraints are used to maintain feasible traffic-management decisions.

3.5 OPTIMIZATION DESIGN

The optimization design converts traffic-management decisions into a mathematical optimization problem. The decisions are represented using binary variables and formulated as a Quadratic Unconstrained Binary Optimization (QUBO) problem.

3.5.1 Decision Variable Formulation

Each possible traffic-management action in the Quantum Traffic Optimizer is represented using a binary decision variable x_i , where $x_i = 1$ indicates that the corresponding action is selected and $x_i = 0$ indicates that it is not selected. This representation allows different types of traffic-management actions to be incorporated into a single optimization model. The decision variables are grouped according to their purpose, including route-diversion decisions, signal-extension decisions, and temporary traffic-restriction decisions.

For the current implementation, the optimization model contains 18 binary decision variables. The first four variables represent route-diversion decisions associated with the identified traffic corridors, variables x_5 to x_{14} represent signal-extension decisions for selected intersections, and variables x_{15} to x_{18} represent temporary-restriction decisions associated with the traffic corridors.

This binary representation enables the optimizer to evaluate different combinations of traffic-management actions as possible solutions to the pre-event planning problem.

3.5.2 Objective Function

The objective function defines the criteria used to evaluate different combinations of traffic-management decisions. In the Quantum Traffic Optimizer, the objective is designed around traffic performance measures associated with the anticipated event scenario. These measures include factors such as traffic queues, expected travel time, and the ability of the

network to accommodate the anticipated vehicle demand. A solution is therefore evaluated based on how effectively the selected decisions support the intended traffic-management objectives.

The objective function assigns mathematical contributions to the selected decision variables and their interactions. Beneficial decisions can contribute positively to the overall objective, while undesirable combinations can be discouraged through appropriate terms in the optimization model. By combining these effects into a single objective function, different possible traffic-management plans can be compared using a common numerical measure. The optimization process then searches for a binary configuration that provides a suitable objective value while satisfying the defined traffic-management requirements.

3.5.3 Constraint Formulation

Constraints ensure that the generated traffic-management plan remains within predefined operational limits. Signal timing, incompatible actions, and corridor-related restrictions can be represented through penalty terms. The current signal configuration uses a green-time range of 30–60 seconds and a yellow interval of 5 seconds.

3.5.4 QUBO Formulation

The Quadratic Unconstrained Binary Optimization (QUBO) formulation converts the traffic-management problem into a mathematical form consisting of binary variables, linear terms, and quadratic interaction terms. The general QUBO objective can be expressed as:

$$\min_x f(x) = x^T Q x$$

where x represents the vector of binary decision variables and Q represents the QUBO matrix. Each element of the matrix represents the contribution of an individual decision or the interaction between two decisions. This formulation allows the relationships between route, signal, and restriction decisions to be represented within a single mathematical model.

Since the original traffic-planning problem may contain explicit constraints, penalty terms are incorporated into the QUBO formulation to discourage solutions that violate the required conditions. The resulting formulation combines the traffic objective and constraint penalties into a single unconstrained binary objective. This provides a standardized representation of the problem that can subsequently be transformed into a form suitable for quantum optimization.

3.5.5 QUBO Matrix Construction

After defining the decision variables, objective terms, and constraint penalties, the corresponding coefficients are arranged into a QUBO matrix. For the current implementation, the 18 binary decision variables result in an 18×18 QUBO matrix. The diagonal elements represent the individual contributions of the corresponding decision variables, while the off-diagonal elements

represent interactions between pairs of decisions. These interactions are important because selecting one traffic-management action can influence the effect or suitability of another action.

The constructed QUBO matrix provides the complete numerical representation of the optimization problem and acts as the input to the subsequent quantum-optimization stage. The matrix is stored and analysed as part of the optimization output, allowing the formulation to be inspected independently from the solver. Once constructed, the QUBO representation can be transformed into the required quantum formulation and used to obtain an optimized binary solution representing the selected traffic-management decisions.

3.6 QUANTUM OPTIMIZATION ARCHITECTURE

The quantum optimization architecture provides the computational stage for solving the binary traffic-management problem. The QUBO formulation is converted into a quantum-compatible representation and processed using a QAOA-based optimization approach. The resulting bitstring is decoded into traffic-management actions and can subsequently be applied to the SUMO simulation.

3.6.1 Qubit Representation

Each binary decision variable is associated with a qubit. The computational states $|0\rangle$ and $|1\rangle$ correspond to the unselected and selected states of a traffic-management action. Therefore, the 18-variable formulation is represented using 18 conceptual qubits, and measurement produces an 18-bit decision string.

3.6.2 Ising Model Conversion

The QUBO formulation is transformed into an equivalent Ising model before being represented in a quantum optimization framework. The Ising model uses spin variables, conventionally represented as $s_i \in \{-1, +1\}$, instead of binary variables $x_i \in \{0, 1\}$. The relationship between the two representations is commonly expressed as:

$$x_i = \frac{1+s_i}{2}$$

or equivalently,

$$s_i = 1 - 2x_i.$$

This transformation allows the binary QUBO problem to be expressed in a form based on interactions between spin variables. The resulting Ising Hamiltonian represents the same optimization objective while providing a formulation that is compatible with quantum optimization methods. The conversion therefore acts as a mathematical bridge between the traffic-management QUBO model and the quantum representation used in the optimization process.

3.6.3 QAOA-Based Optimization

The Quantum Approximate Optimization Algorithm (QAOA) uses a cost Hamiltonian derived from the optimization problem and a mixer Hamiltonian to explore possible solutions. Adjustable parameters are optimized to increase the probability of obtaining low-cost solutions. The resulting measurements produce candidate bitstrings, which are evaluated using the QUBO objective function. The selected bitstring is decoded into the corresponding traffic-management decisions.

3.6.4 Quantum and Classical Optimization Comparison

A classical optimization approach is used as a reference for evaluating the quantum formulation. Both approaches operate on the same decision variables and optimization problem, allowing comparison of objective values, selected decisions, computational runtime, and traffic-performance measures.

The comparison provides an experimental basis for examining the applicability of quantum optimization. Since the current experiments use software-based quantum simulation or classical execution, the results do not represent the performance of a future large-scale fault-tolerant quantum computer.

3.7 SYSTEM WORKFLOW

The complete workflow integrates event-data processing, SUMO microscopic simulation, Digital Twin visualization, and quantum optimization into a closed-loop traffic-planning process. Event information, road-network data, and expected traffic demand are first used to construct the traffic scenario. The prepared road network and traffic demand are then loaded into SUMO, where the traffic environment is simulated at the microscopic level.

The Python backend communicates with SUMO through TraCI and obtains traffic information such as vehicle movement, congestion, queues, and signal states. This traffic state is used for optimization and evaluation. The traffic-management decisions are represented using 18 binary variables and formulated as a QUBO problem. The QUBO is processed by the QAOA-based optimization module to generate candidate solutions.

The selected optimized decisions are decoded and mapped to the corresponding SUMO network elements. Through TraCI, the decisions are applied to the simulation, allowing the updated traffic condition to be evaluated. The updated simulation state is then transmitted through WebSocket communication to the Three.js Digital Twin, which provides a 3D representation of the simulated environment and displays relevant traffic and optimization information.

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGY

The proposed system implements a simulation-driven Digital Twin and hybrid quantum optimization framework for pre-event traffic planning. The road network and traffic demand are represented in SUMO, where vehicle and pedestrian movement is simulated at microscopic level. The Python controller communicates with SUMO through TraCI and collects the required traffic information. Based on the selected event scenario, traffic demand and corridor conditions are used to formulate an optimization problem with 18 binary decision variables. The resulting QUBO is converted into an Ising representation and solved using QAOA through a quantum simulation backend. The obtained bitstring is decoded into route diversion, signal timing, and temporary restriction decisions. These decisions are mapped to the corresponding SUMO edges and traffic signals through TraCI, after which the updated traffic state is observed in the Digital Twin.

4.2 SUMO-BASED DIGITAL TWIN IMPLEMENTATION

The Digital Twin is implemented as a simulation-driven representation of the stadium traffic environment using SUMO. The SUMO microscopic simulator represents the road network, vehicles, pedestrians, traffic signals, and elevated railway movement. The simulation advances in 0.5-second steps and provides the traffic state required for optimization and evaluation.

A Python-based controller communicates with SUMO through TraCI. It manages the simulation scenario, retrieves vehicle and pedestrian states, monitors congestion-related measures, and applies the optimized traffic-management decisions. The implementation supports Normal Day and Event Day scenarios, with event conditions represented through increased vehicle and pedestrian demand.

The Digital Twin follows a three-layer structure consisting of the SUMO simulation layer, Python controller layer, and visualization layer. SUMO provides the simulated traffic state, the Python backend processes the state and communicates with SUMO, and the Three.js frontend displays the updated traffic environment. The backend transmits simulation information through WebSocket communication for real-time visualization.

The implemented traffic model includes cars, two-wheelers, buses, pedestrians, and elevated MRTS movement. Vehicle behaviour is represented using the Intelligent Driver Model (IDM), while pedestrian movement and railway operation are represented using the defined simulation models. Event-day demand is generated using the specified temporal arrival pattern, including pre-event, arrival-surge, peak, and post-event phases.

The Digital Twin operates as a closed loop. The baseline traffic state is generated in SUMO, relevant traffic conditions are passed to the optimization module, and the resulting route, signal, and restriction decisions are returned to SUMO through TraCI. The updated simulation state is then visualized and evaluated using traffic-performance measures such as travel time and queue length.

4.3 DETAILED QUANTUM OPTIMIZATION PROCESS

4.3.1 Decision Variable Encoding

The optimization model uses 18 binary variables to represent the available traffic-management decisions. The variables are grouped according to their function. Variables $x_1x_1-x_4x_4$ represent route-diversion decisions for the selected traffic corridors, $x_5x_5-x_{14}x_{14}$ represent signal-timing decisions for intersections J1–J10, and $x_{15}x_{15}-x_{18}x_{18}$ represent temporary traffic-restriction decisions. A value of 1 activates the corresponding decision, while 0 leaves it inactive. This encoding provides a compact representation of the possible pre-event traffic-management actions.

4.3.2 Objective and Penalty Terms

The optimization model uses 18 binary variables to represent the available traffic-management decisions. The variables are grouped according to their function. Variables $x_1x_1-x_4x_4$ represent route-diversion decisions for the selected traffic corridors, $x_5x_5-x_{14}x_{14}$ represent signal-timing decisions for intersections J1–J10, and $x_{15}x_{15}-x_{18}x_{18}$ represent temporary traffic-restriction decisions. A value of 1 activates the corresponding decision, while 0 leaves it inactive. This encoding provides a compact representation of the possible pre-event traffic-management actions.

4.3.3 QUBO Construction

After defining the decision variables, objective terms, and penalty terms, the complete traffic optimization problem is converted into a Quadratic Unconstrained Binary Optimization (QUBO) formulation. The general form of the QUBO problem is expressed as $f(x) = x^T Q x$, where x is the binary decision vector and Q is the QUBO coefficient matrix. For Proj_v2, the decision vector contains the 18 binary variables representing route diversions, signal extensions, and temporary restrictions. The diagonal elements of the matrix represent the individual contribution of each decision variable, while the off-diagonal elements represent interactions between pairs of decisions.

The objective and penalty components are combined to produce a single unconstrained binary optimization function. This formulation allows the traffic-management problem to be expressed in a form suitable for quantum optimization. The resulting QUBO matrix has a dimension of 18

$\times 18$, corresponding to the 18 decision variables. The matrix is generated programmatically and stored for further processing, including conversion to the Ising representation and preparation of the quantum optimization problem.

4.3.4 Quantum Circuit and Qubit Mapping

The binary decision variables in the QUBO formulation are mapped to quantum bits (qubits) for the quantum optimization stage. Each binary variable x_i is associated with a qubit whose computational basis states represent the two possible values of the corresponding decision: $|0\rangle$ represents $x_i = 0$, and $|1\rangle$ represents $x_i = 1$. Since Proj_v2 contains 18 binary decision variables, the conceptual quantum formulation uses 18 qubits, with each qubit corresponding to one traffic management decision. A complete measurement of these qubits therefore produces an 18-bit binary string representing one possible traffic management plan.

Before applying QAOA, the QUBO formulation is converted into an equivalent Ising representation using binary-to-spin variable transformation. The binary variable can be related to the spin variable through $x_i = (1 - s_i)/2$, where $s_i \in \{1, -1\}$. The resulting Ising formulation is represented using a cost Hamiltonian, which provides the basis for constructing the quantum optimization circuit. In the QAOA framework, the cost Hamiltonian represents the traffic optimization objective, while a mixing Hamiltonian is used to explore different combinations of qubit states.

4.3.5 QAOA Optimization Process

The Ising formulation is used to construct the cost Hamiltonian for QAOA. The quantum circuit alternates between the cost and mixer operators for the selected QAOA depth. Classical parameter optimization is used to adjust the circuit parameters and minimize the estimated objective value. In the current implementation, QAOA is executed using a classical quantum simulator; therefore, the reported execution time represents simulated quantum computation rather than execution on physical quantum hardware.

4.3.6 Optimal Bitstring Extraction

After the quantum optimization stage, the measured qubit states are converted into binary bitstrings corresponding to the original traffic-management decision variables. Each position in the bitstring maintains the same ordering used during QUBO construction. Therefore, the first four bits correspond to route-diversion variables x_1-x_4 , the next ten bits correspond to signal-extension variables x_5-x_{14} , and the final four bits correspond to temporary traffic-restriction variables $x_{15}-x_{18}$. A bit value of 1 indicates that the corresponding action has been selected, while a bit value of 0 indicates that it has not been selected.

The selected bitstring is subsequently decoded into meaningful traffic-management decisions. The active route-diversion bits identify the corridors considered for diversion, the active signal-extension bits identify the intersections requiring signal adjustments, and the active restriction bits identify the corridors where temporary restrictions are considered. The decoded decisions are then used to construct the final pre-event traffic management plan, which can be evaluated using the traffic-performance measures defined in the system. This final decoding step connects the mathematical optimization output with practical actions that can be considered by traffic authorities during pre-event planning.

4.4 TRAFFIC MANAGEMENT DECISION GENERATION

4.4.1 Route Diversion Decisions

The first four decision variables, $x_1x_1-x_4x_4$, represent route-diversion actions for the four identified traffic corridors. When a variable is activated, the corresponding corridor decision is mapped to its predefined SUMO edges. The decoded route decisions are then applied through the TraCI controller so that their effect can be evaluated within the Digital Twin.

4.4.2 Signal Timing Decisions

Variables $x_5x_5-x_{14}x_{14}$ represent signal-timing decisions for intersections J1–J10. The selected binary values are decoded and mapped to the corresponding traffic-light identifiers in the SUMO network. The implementation allows the optimized timing values to be transferred to the simulation through TraCI. This connects the quantum optimization result with the actual signal-control representation of the Digital Twin.

4.4.3 Temporary Traffic Restriction Decisions

Temporary traffic restrictions are represented by the final four binary variables, $x_{15}-x_{18}$, with each variable associated with one of the selected traffic corridors. A value of 1 indicates that a temporary restriction is included as a candidate decision for the corresponding corridor, while a value of 0 indicates that the restriction is not selected. These restrictions can represent pre-planned measures such as limiting certain movements or controlling access through a corridor during the anticipated event-traffic period.

The restriction decisions are evaluated together with route diversions and signal adjustments rather than being considered independently. This is important because restricting one corridor may influence traffic demand on neighbouring routes and intersections. After the optimal bitstring is decoded, the active restriction variables are identified and incorporated into the final traffic-management plan. The restrictions are intended to be implemented only where operationally appropriate and are subject to the authority's assessment of safety, accessibility, emergency movement, and actual event-day conditions.

4..4.4 Final Pre-Event Traffic Management Plan

The final pre-event traffic management plan is generated by combining the three categories of decisions obtained from the optimized solution: route diversions, signal timing adjustments, and temporary traffic restrictions. The decoded bitstring provides the selected actions, while the corresponding corridor, intersection, and traffic information provides their practical meaning. Instead of presenting the optimization result only as a mathematical binary string, the system converts it into an interpretable set of traffic-management actions that can be reviewed before the event.

The resulting plan provides traffic authorities with a structured representation of how the anticipated event traffic can be managed across the selected road network. The plan identifies the corridors considered for diversion or restriction, the intersections selected for signal adjustments, and the corresponding operational settings. It can then be evaluated using measures such as travel time, queue levels, and throughput and compared with a reference classical optimization solution. Thus, the final output of the framework is a decision-support plan for pre-event traffic preparation rather than an autonomous system for controlling traffic during the event.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 EXPERIMENTAL SET UP

The experimental setup was developed to evaluate the proposed Quantum Traffic Optimizer in a pre-event traffic planning scenario. An IPL match at the M.A. Chidambaram Stadium in Chennai was selected as the representative public-event scenario because such events can generate a concentrated increase in vehicle demand around the venue and its surrounding road network. The study area was defined around the stadium using a network radius of approximately 1200 m, with relevant traffic corridors identified within the surrounding region. The road network was obtained from OpenStreetMap and processed using OSMnx, resulting in a network containing approximately 750 intersections and 1985 road segments. Four major traffic corridors—Anna Salai, Triplicane High Road, Wallajah Road, and Kamarajar Salai—were identified for traffic-management decisions.

The traffic demand was configured using an expected event demand of 9515 vehicles distributed across the selected entry/intersection points. The demand values used in the scenario were $J1 = 1200$, $J2 = 1447$, $J3 = 600$, $J4 = 1100$, and $J5 = 450$ vehicles, with the remaining network used to represent the surrounding traffic movement. The experiment also incorporated the expected temporal arrival behaviour of event vehicles so that the traffic scenario represented a gradual buildup of demand rather than assuming that all vehicles reached the venue simultaneously. Signal-management constraints were defined with a minimum green time of 30 seconds, a maximum green time of 60 seconds, and a yellow time of 5 seconds. These inputs were used to establish the operational conditions under which the optimization was performed.

For the optimization experiment, 18 binary decision variables were defined, consisting of four route-diversion variables, ten signal-extension variables corresponding to $J1$ – $J10$, and four temporary-restriction variables. These variables were used to construct an 18×18 QUBO matrix containing the objective and interaction terms of the traffic-management problem. The QUBO formulation was subsequently prepared for the quantum optimization stage through binary-to-Ising conversion and qubit representation. A classical optimization approach was also considered using the same decision variables and objective formulation to provide a common reference for comparison. The resulting solutions were decoded into route, signal, and restriction decisions and evaluated using traffic-performance measures such as travel time, queue levels, and throughput.

5.2 PERFORMANCE AND RESULT

The proposed pre-event traffic optimization framework was benchmarked using five scenario configurations that could be characterized as clear weather, heavy rain, VIP movement, roadwork, and a combined scenario that incorporated multiple constraints. The framework was

evaluated using three optimization approaches including greedy optimization, classical optimization, and the QAOA-based optimization approach.

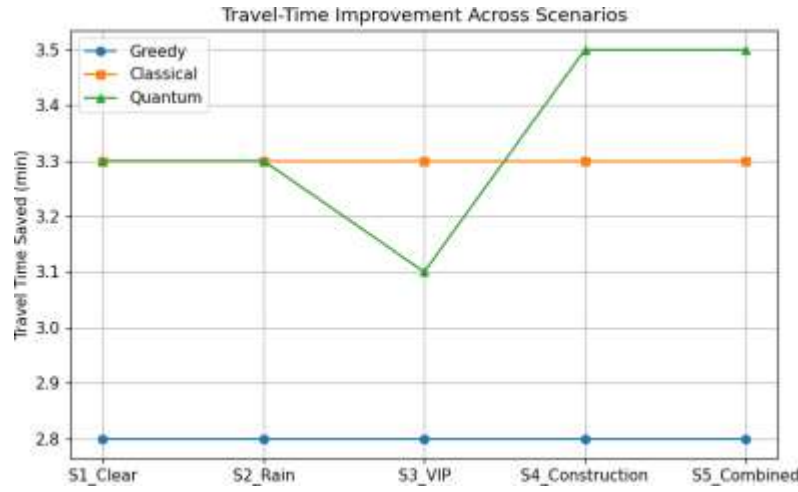


Figure 2: Travel Time Improvement Across Scenarios

The results on the travel-time demonstrate that the greedy approach yielded a constant saving time of 2.8 minutes for all the cases. The classical optimization approach, on the other hand, resulted in a saving time of 3.3 minutes for all the scenarios. The QAOA-based approach resulted in 3.3, 3.1, and 3.5 minutes of travel-time saving for the clear and heavy-rain, VIP, and construction and combined scenarios, respectively.

Therefore, the QAOA-based approach matched the classical optimization approach in terms of the travel-time saving for the clear and heavy-rain scenarios. For the VIP scenario, the saving time was 0.2 minutes less than the saving time yielded by the classical optimization approach. However, for the construction and combined scenarios, the saving time of the QAOA-based approach was 3.5 minutes as compared to 3.3 minutes of the classical optimization approach. Therefore, the combined scenario represents an increase of 0.2 minutes of observed saving time for the QAOA-based approach over the classical optimization approach.

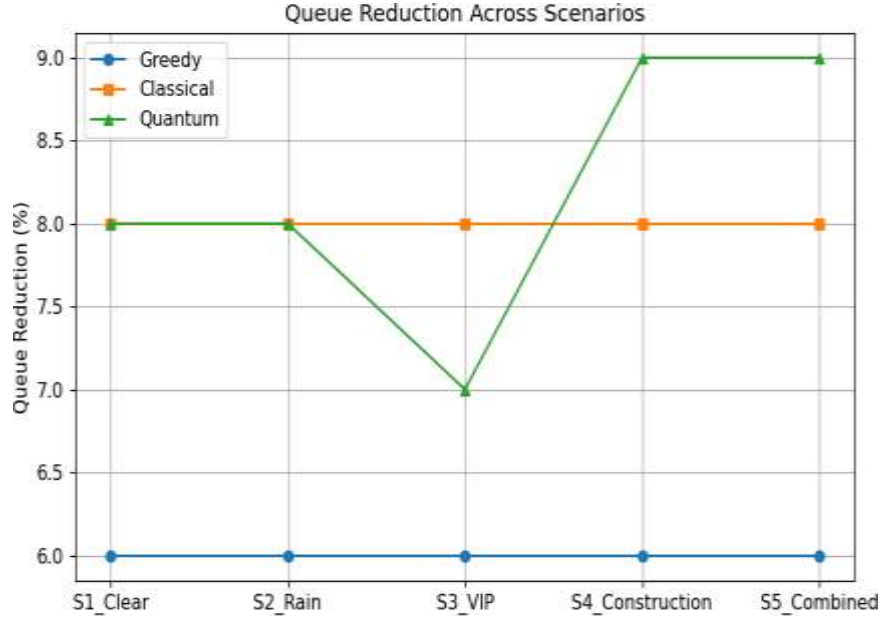


Figure 3 : Queue Reduction Across Scenarios

A similar trend was observed with respect to the reduction in queue. The greedy approach accomplished a 6% reduction of all cases, whereas the classical approach accomplished an 8% reduction overall. The QAOA-based approach accomplished an 8% reduction on clear and heavy-rain conditions, 7% reduction on the VIP scenario, and 9% reduction on construction and combined cases.

The QAOA-based approach therefore matched the classical approach in terms of the reduction in the queues under clear and heavy-rain conditions, and it outperformed the classical approach on the construction and combined scenarios. On the combined scenario, the QAOA-based approach accomplished a 9% reduction in the queues compared to 8% for the classical approach. These results suggest that the QAOA-based formulation maintained a comparable level of traffic-management performance when individual and combined operational constraints are introduced.

The scenario-level travel-time results further illustrate the consistency of the optimization approaches. The classical approach was steady at 3.3 minutes in every scenario, while the QAOA-based approach reacted to the scenario properties. Its saving reduced to 3.1 minutes for the VIP condition but raised to 3.5 minutes under construction or the combined scenario. This implies that the QAOA-based approach was sensitive enough to the modifications made to the optimization scenario to respond accordingly rather than produce the same traffic-management result for any condition.

The average result across all scenarios provides a general comparison perspective. The greedy, classical, and QAOA-based approaches achieved 2.80, 3.30, and 3.34 minutes of average travel-time saving, respectively. Correspondingly, the approaches produced 6.0%, 8.0%, and 8.2% decreases in average queue. Thus, compared to the greedy baseline, the QAOA-based approach improved the average travel-time saving by 19.286% and the average queue reduction by 36.667%. The classical approach corresponded to 17.857% improvement in average travel-time saving and 33.333% improvement in average queue reduction. The average results are directly reported from the experimental efficiency report.

The average results suggest that the QAOA-based approach produced the best traffic-performance result in terms of the average travel-time saving and average queue reduction compared to the other two approaches in the current experimental evaluation. Its improvement over the classical approach was minor, amounting to 0.04 minutes in average travel-time saving and 0.2 percentage points in average queue reduction. Therefore, the main finding resulting from the comparison is that the QAOA-based approach achieved the traffic-performance result comparable to the classical approach rather than outperforming it.

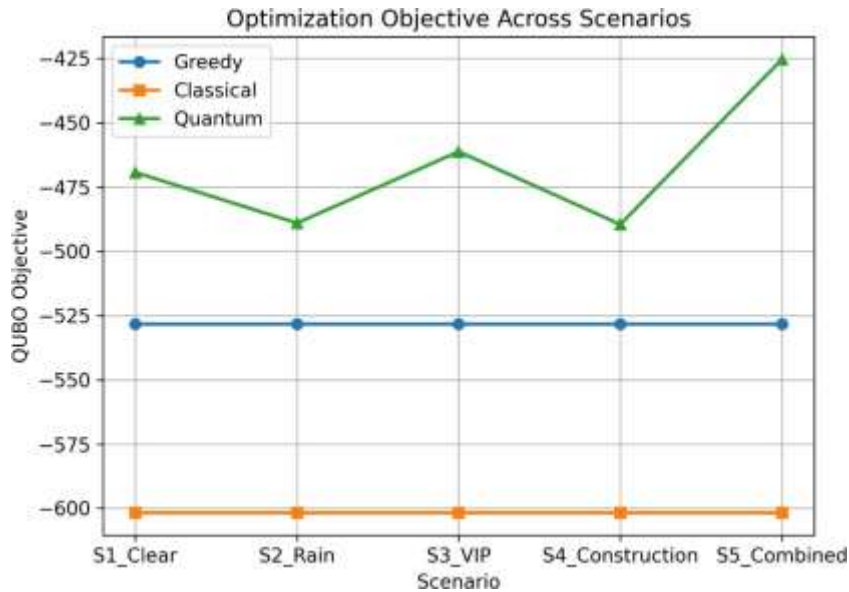


Figure 4 : Optimization Objective Across Scenarios

The QUBO objective values were also investigated to assess the behaviour of the mathematical optimization formulation. The greedy and classical objective values were constant at five scenarios at approximately -528.28 and -601.68, respectively. On the other hand, the QAOA-based objective changed with the scenario, taking values of -469.25 for the clear

scenario, -489.00 for heavy rain, -461.20 for the VIP scenario, -489.50 for construction, and -425.25 for the combined scenario.

The average QUBO objective values were -528.28 for greedy, -601.68 for classical and -466.84 for the QAOA-based approach. Since the QUBO objective represents the objective of the mathematical optimization problem, these values have to be considered separately to traffic-performance indicators, as travel-time saving and queue reduction. In the current formulation, the classical method achieved a numerically lower QUBO objective, while the QAOA-based method achieved competitive traffic-performance outcomes. This shows that the optimization-objective value and downstream traffic-performance indicators are related but non-interchangeable evaluation metrics.

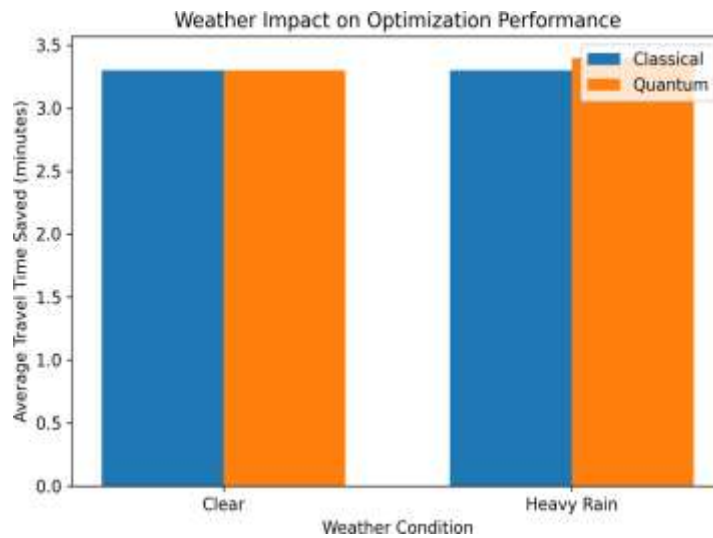


Figure 5: Weather Impact on Optimization Performance

The impact of weather conditions was analyzed by comparing the clear and heavy-rain scenarios. For the clear setting, the classical and QAOA-based approaches both resulted in 3.3 minutes of travel-time saving. The same was observed for the heavy rain condition, where the classical approach produced 3.3 minutes of travel-time saving, and the QAOA-based method produced 3.3 minutes as well. Correspondingly, both the approaches reduced the queues by 8% each in both the settings.

Based on the weather-specific analysis, it can be concluded that the QAOA-based approach was able to maintain the same level of traffic-performance as that of the classical approach when the scenario was switched from clear weather to heavy rain. This provides evidence that the optimization formulation was able to remain stable under the environmental change that was encapsulated within the experimental scenarios.

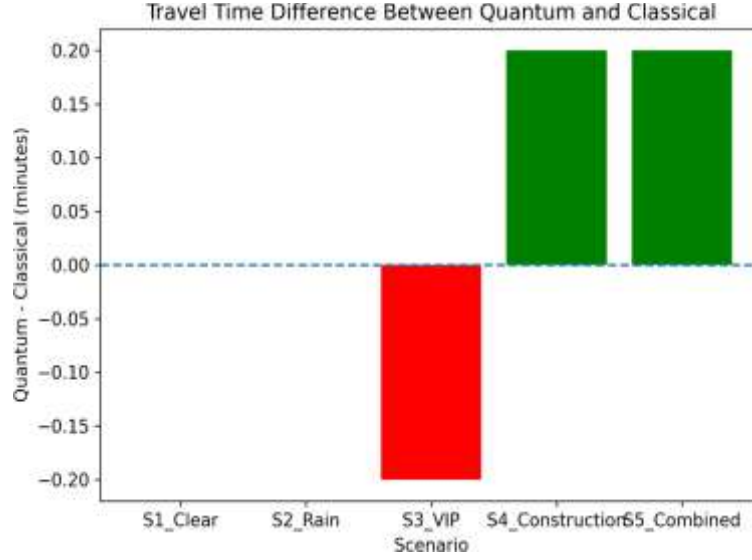


Figure 6: Travel Time Difference Between Quantum and Classical Algorithm

The direct comparison between the QAOA-based and classical approaches further illustrates the results' scenario dependence. The travel-time difference was approximately zero for the clear and heavy-rain scenarios, -0.2 minutes for the VIP scenario, and $+0.2$ minutes for both the construction and combined scenarios. Thus, the QAOA-based method did not systematically outperform or underperform the classical approach in all scenarios. Rather, its relative performance varied according to the constraints embodied in the scenario.

This behaviour is particularly relevant to the combined scenario, which embodies multiple simultaneous constraints. In this case, the QAOA-based approach achieved 3.5 minutes of travel-time saving and a 9% queue reduction, compared with 3.3 minutes and 8%, respectively, for classical optimization. Thus, the combined scenario produced the highest observed traffic-performance values for the QAOA-based approach in the present evaluation.

The experimental results show that the proposed QAOA-based formulation successfully operated across five different pre-event traffic scenarios and produced traffic-management performance close to that of the classical optimization approach. Its average travel-time saving was 3.34 minutes compared with 3.30 minutes for classical optimization, while its average queue reduction was 8.2% compared with 8.0% for the classical approach. Relative to the greedy baseline, the QAOA-based method achieved 19.286% improvement in average travel-time saving and 36.667% improvement in average queue reduction.

At the scenario level, the QAOA-based method matched the classical method under clear and heavy-rain conditions, produced slightly lower traffic-performance values under the VIP scenario, and produced higher observed travel-time saving and queue reduction under both construction and combined conditions than did the classical method. The combined scenario

yielded 3.5 minutes of travel-time saving and 9% queue reduction using the QAOA-based approach, compared with 3.3 minutes and 8% for the classical approach.

The computational analysis revealed a different trade-off than that observed at the scenario level. Specifically, the QAOA-based implementation required approximately 18.03 seconds on average than did the 2.55 seconds required for classical optimization. The overall efficiency score was also lower for the QAOA-based method than for the classical method. These results are consistent with the fact that the present QAOA formulation was evaluated using classical simulation rather than physical quantum hardware.

Taken together, the results highlight the feasibility and competitive traffic-performance behaviour of the QAOA-based formulation for the proposed pre-event traffic optimization problem. Furthermore, the experiments do not establish a general quantum computational advantage, since the present implementation is simulation-based and the classical approach remains more efficient in terms of runtime and composite efficiency. However, the results show that the QAOA-based formulation can produce traffic-management solutions with performance comparable to classical optimization and can achieve higher observed traffic-performance values under the construction and combined scenarios.

The SUMO-based Digital Twin module was evaluated to verify its ability to represent the Chepauk traffic environment, simulate multiple modes of transport, apply traffic-management constraints, and visualize the resulting traffic state. The developed interface provides separate Classical Baseline and Quantum Optimized simulation views, allowing the same traffic scenario to be observed before and after optimization. The Digital Twin represents cars, buses, two-wheelers, pedestrians, and local trains within the simulated road network, while traffic signals and road conditions are managed through the SUMO–TraCI interface.

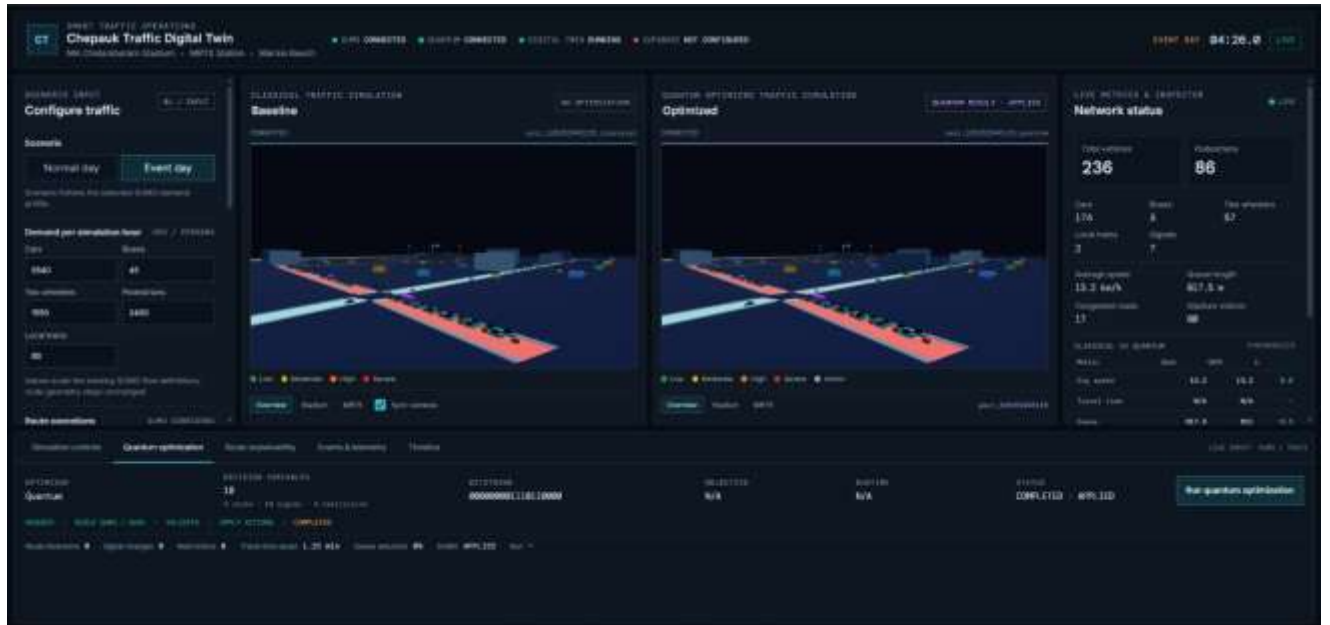


Figure 7: SUMO-Based Chepauk Traffic Digital Twin with Classical and Quantum Simulation.

As shown in Figure , the Event Day simulation provides a three-dimensional representation of the Chepauk road network. The interface displays the movement of different transport modes along the simulated roads while using different visual indicators to represent traffic conditions. The live metrics panel provides information about the current network state, including vehicle categories, pedestrians, local trains, traffic signals, average speed, queue length, and stadium visitors. The same environment is represented in both the classical and quantum views, which allows the effect of optimization decisions to be observed without changing the underlying road geometry.

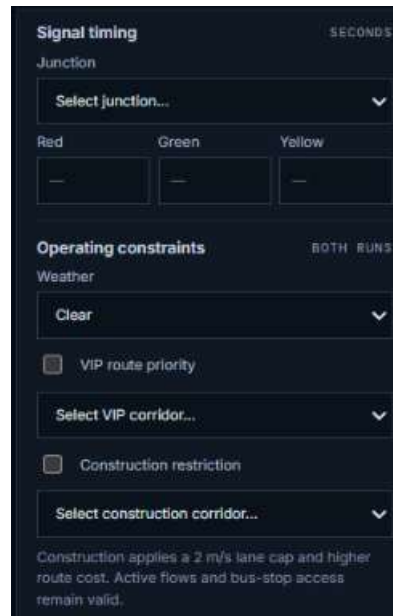


Figure 8: *Traffic Signal and Event Constraint Configuration in the Digital Twin*

Through **SUMO–TraCI integration**, traffic-management decisions can be transferred to the running simulation and reflected in the Digital Twin visualization. The system successfully mapped control actions to the corresponding road and junction elements, allowing route conditions and signal timings to be modified during simulation. Overall, the results demonstrate that the developed Digital Twin provides an interactive environment for **traffic simulation, scenario configuration, control-action application, and visualization** for the proposed traffic-management system..

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The proposed Quantum Traffic Optimizer presents a pre-event traffic planning framework for managing anticipated vehicle movements during large public events. The M.A. Chidambaram Stadium, Chennai, was used as the experimental scenario, incorporating real-world road-network information, expected vehicle demand, temporal arrival behaviour, traffic corridors, intersections, and operational constraints. The traffic-management problem was formulated using binary decision variables for route diversions, signal timing adjustments, and temporary traffic restrictions. These decisions were represented as an 18-variable QUBO problem with an 18×18 QUBO matrix, followed by qubit encoding and Ising-model conversion for quantum optimization. The resulting binary solution was decoded into practical traffic-management actions. Overall, the framework provides a systematic approach for generating and evaluating alternative traffic-management plans before a public event. Traffic-performance measures such as travel time, queue levels, and throughput can be used to evaluate the resulting plans. The work demonstrates the applicability of QUBO-based quantum optimization as a decision-support approach for pre-event traffic planning, while final implementation remains subject to operational validation by traffic authorities.

The results confirm that the SUMO–TraCI Digital Twin successfully provides the simulation foundation for the proposed traffic-management system. It represents the Chepauk environment as a multimodal traffic network containing roads, junctions, traffic signals, vehicles, pedestrians, and local trains, while also supporting event-specific conditions such as weather, VIP movement, and construction. The side-by-side classical and quantum simulation views enable traffic states to be observed and compared within the same virtual environment. The integration of SUMO, TraCI, the Python backend, and the Three.js interface therefore establishes a complete simulation-to-visualization pipeline in which traffic conditions can be configured, optimization decisions can be applied, and the resulting traffic state can be visualized. Overall, the Digital Twin provides a practical environment for testing and demonstrating quantum-assisted traffic-management strategies before their potential application to real-world traffic systems.

6.2 FUTURE WORK

Future work can improve the traffic-demand model by incorporating larger sets of real-world traffic observations, historical event data, vehicle-origin information, parking demand, public-transport usage, and more detailed time-dependent arrival patterns. These additions can provide a more detailed representation of event-related traffic demand. The quantum optimization component can be extended to actual quantum hardware and evaluated using different QAOA circuit depths and parameter-optimization strategies. Larger road networks and additional decision variables can also be studied to examine scalability. Further constraints, including emergency-vehicle access, road capacity, pedestrian movement, parking restrictions, one-way traffic, and neighbouring signal coordination, can be incorporated. Finally, the framework can be extended to different event types and locations, including concerts, festivals, sports competitions, and religious gatherings. Future experiments can also use larger benchmark instances, multiple solver configurations, and repeated runs to provide a more comprehensive comparison between quantum and classical optimization approaches.

Future work will focus on improving the SUMO–TraCI Digital Twin into a more realistic and scalable traffic-management platform. Real-time traffic data from CCTV cameras, GPS, roadside sensors, and public transportation systems can be integrated to continuously update the simulated traffic conditions. The Digital Twin can also be expanded to support larger road networks and more detailed modelling of vehicles, pedestrians, public transport, and event-related crowd movement. Further development can include advanced traffic-signal control, emergency-vehicle priority, temporary road closures, parking conditions, and dynamic VIP routes. The current system can also be extended with cloud-based data storage to maintain historical simulation and optimization results for long-term analysis.

Future studies should evaluate the system across a larger number of traffic scenarios and repeated simulation runs to obtain statistically reliable performance measures. The QAOA-based optimization can also be tested on larger problem sizes and, where available, on actual quantum hardware. Ultimately, the Digital Twin can be developed into a real-time closed-loop system, where live traffic conditions automatically trigger optimization, the resulting decisions are applied through TraCI, and the updated traffic state is continuously visualized and fed back into the optimization process.

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Hybrid Quantum Traffic Optimization Framework Using Digital Twins for Major Public Events.

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Abstract

Vehicles and pedestrians from major public events like sports, music, festivals, and faith meetings create massive numbers quickly, stressing transport systems. Traffic rises during these events causing bad congestion, poor signal timing, long delays, and hard flow maintenance. Even though Digital Twins monitor city traffic well, many tools focus on regular conditions and lack event-specific help. At the same time, traditional traffic improvement techniques fail to meet the massive, shifting, and multiple-goal characteristics found within transport networks during significant public gatherings. Thus, this paper proposes a Hybrid Quantum Traffic Optimization Framework Using Digital Twins for Major Public Events. A new Scheme for Traffic Simulation Setting and Planning Event Optimization via Digital Twin-based Traffic Simulation Environment and Hybrid Quantum-classical Optimization .This paper suggests a scheme joining Digital Twin-based traffic simulation setting with quantum-classical hybrid optimization to model, examine, and improve traffic states ahead of planned gatherings. Digital Twin models and produces the virtual setting of the transport network allowing road facilities, vehicle movement, traffic data, and

other relevant parameters, plus utilizes QUBO formulation to build a hybrid quantum-classical optimizer capable of modeling and improving traffic control steps regarding path diversion, traffic signal settings, and short-term road shutdowns. Transport officials may assess and choose the superior traffic control plan among various options employing the suggested Digital Twin-based traffic simulation setting within a simulator-based setting.

Keywords - *QUBO , QAOA , Ising Hamiltonian , Corridor analysis , Digital Twin , SUMO , TraCI , Non Homogeneous poisson process.*

1.INTRODUCTION

Large public gatherings involve a number of vehicles aggregating together leading to extensive congestion and extended travel distances. Even though the traffic is anticipated, its management is based on intuition, experience. Such techniques prove to be efficient but suboptimal therefore requiring smart flexible solutions heading for quantum computing providing a hope for addressing difficult combinatorial tasks more simply than traditional computers based on principles such as superposition hence identifying best answers to huge traffic control issues[9].

In classic systems, the basic unit of information used in computing and data transmission is a bit which can either be 0 or 1. On the contrary, quantum computing uses the qubit as its fundamental information. By definition, superposition dictates that qubits can be in multiple states at the same time i.e. they can be in both zero and one simultaneously. Superposition enables quantum computers to process an exponential number of possibilities at the same time on a large scale.

A Digital Twin is used to design a traffic environment, where traffic conditions can be compared to alternative optimized scenarios. In this way, a Digital Twin environment allows for both what-if analysis and assessment of performance relative to optimized alternatives. At the same time, quantum approaches formulate the traffic management problem as a combinatorial optimization problem. Since both route and traffic-control variables can make optimization a NP-hard task [13], the quadratic unconstrained binary optimization (QUBO) was chosen to encode decisions as binary variables and encode objective functions and penalties in a combined energy function. In this way, the binary variables indicate diversions for certain routes, extensions of certain junctions, and temporary restrictions of selected traffic routes.

The existing research has limited research considering the application of quantum optimization algorithms to event-based traffic management scenarios that require multiple coordinated decisions. The work by Yarkoni et al. [3] showed the ability of quantum computers to perform vehicle-route optimization for the Web Summit based on real-time traffic information, without considering coordinated optimization of multiple traffic-control decisions during the event. The use of quantum annealing for optimization has seen issues with scalability, connectivity, and other practical considerations for traffic management applications [6]. This motivates proposing a pre-event coordinated optimization framework that combines Digital Twin-based modeling and quantum-classical optimization of connected traffic-management decisions.

While Digital Twins have been studied extensively for quantum-optimized coordinated event-specific traffic management [2][7][8] simultaneous optimization of route diversions, signal extensions, and temporary

restrictions have received limited attention, with few demonstrations in practice [5][6]. As such, the coordinated optimization of multiple traffic-management decisions for event scenarios is an insufficiently explored area of research.

This work presents a pre-event quantum-classical traffic optimization framework for large public events, which encodes route diversions, signal extensions, and temporary restrictions as binary variables in a single QUBO and optimized using QAOA. A Digital Twin-based urban road network and pre-event synthetic demand is used to benchmark the resulting traffic-management plan against a classical approach, assessing coordinated pre-event planning capabilities, without making claims on quantum speedup or practical operationalization.

2. LITERATURE REVIEW

Yarkoni et al. [3] designed a bus-route management system for the 2019 Web Summit, issued in Lisbon: route selection was translated into a QUBO problem, which was solved on nine buses provided with D-Wave 2000Q processors. It proved the feasibility of the solution, which however was constrained by low levels of quantum resources and system scales. Kumarasamy et al. [2] experimented with an approach based on a Digital Twin and reinforcement-learning to signal control in an 11-intersection network in the MLK Smart Corridor; the approach lowered traffic delays and stops over traditional control, but it was limited to an 11-intersection network and was subject to information transfer delays.

Transportation. Mohammed et al. [6] surveyed quantum annealing applications for transportation such as traffic-signal control, traffic-flow optimization, and vehicle routing [11]. QUBO/Ising formulations were frequently used, but problems with qubit connectivity, hardware limitations, noise, and size persist. Faliagka et al. [8] developed a Traffic Monitoring and What-If analysis Digital Twin framework that takes data from sensors, vehicles, and applications; however, the approach was limited to a Digital Twin architecture, with technology integration, implementation costs, privacy, and infrastructure maturity as identified challenges.

Dribi et al. [5] performed a PRISMA systematic review of 165 papers on intelligent traffic management, found challenges such as scalability, data of different types, interoperability, latency,

privacy, and access to real-world validation. Kumarasamy et al. [7] extended the work to propose a Digital Twin-assisted decentralized multi-agent reinforcement-learning method for the optimization of traffic signal timing in an 11-intersection corridor, finding improvements in Eco-PI compared to actuated control but did not include quantum optimization.

Kim et al. [4] presented quantum annealing for large optimization problems reporting faster execution for some large-scale problems on hybrid quantum annealing but not in transportation-specific context. Prabakaran [1] proposed a FedQ-Twin by combining the concepts of federated learning, quantum-inspired optimization, and Digital Twin simulation of traffic management which needs further investigation of the real-world traffic data and physical infrastructure. In general, the previous work shows a progress of quantum optimization, Digital Twins, and intelligent traffic management[9], but there is still only a little research combined Digital Twin-based traffic simulation with quantum optimization for planning traffic-management strategies for a large public event. This research gap encourages the development of a hybrid quantum-classical framework for investigating the performance of event-specific traffic-management strategies in comparison with the actual traffic system.

3. MATERIALS AND METHOD

3.1 Dataset and Scenario

The assumed event based traffic-management scenario is an IPL match at the M. A. Chidambaram Stadium, with 4 key traffic corridors and a demand level of 9515 vehicles. The demand was artificially simulated based on an assumptions matrix derived from real-world based assumptions, taking into account road network configuration, traffic parameters, route choice options, limitations and signal extension levels. The event intensity, closure level, disruption level and demand multiplier were set at 0.8, 0.3, 0.2 and 1.2 respectively.

3.2 Hardware and Software

This work is implemented by using the Python 3.x, Qiskit 2.1.2, and Qiskit Aer 0.17.1 libraries. For data pre-processing and visualization, various libraries such as NumPy, Pandas, NetworkX, and Matplotlib have been used. The QAOA implementation was done using the statevector simulator of Qiskit Aer library.

3.3 Parameter settings

Parameter settings in the optimization experiment are summarized in Table 2. The classical brute-force approach used the same 18-variable formulation and sampled all $2^{18} = 262,144$ binary bitstrings. Random seed for the classical brute-force optimization was set to 42 and the same seed was used for all approaches as well. The QAOA circuit was evaluated with statevector simulator (without shot-based sampling).

Parameter	Setting
Number of binary decision variables	18
QAOA circuit depth (p)	2
γ values	$0, \pi/4, \pi/2$
β values	$0, \pi/4, \pi/2$
Number of parameter combinations	9
Quantum simulator	Qiskit Aer Statevector Simulator
Random seed	42

Table 1. Parameter Settings Used in the Optimization Experiment

4. Existing Method

An existing approach for event-based traffic optimization is the Quantum Shuttle framework proposed by Yarkoni et al. (2020). The framework addresses vehicle route assignment during large-scale events by formulating the route-selection problem as a Quadratic Unconstrained Binary Optimization (QUBO) problem. Its objective is to assign routes to multiple vehicles while reducing overlap between their paths. The resulting QUBO problem is solved using quantum computing to obtain a suitable combination of vehicle routes.

4.1 Working Principle

Yarkoni et al. suggested the Quantum Shuttle technique for guiding vehicle routes for large events using a combination of transportation data, candidate-route generation, and QUBO-based optimization. The event demand determines routes, vehicle and traffic data determine candidate routes and are then filtered

based on travel times. All remaining routes are encoded as binary variables, optimized with QUBO, and returned for travel [3],[12] .

4.2 Limitations

The Quantum Shuttle framework demonstrated the use of quantum optimization for event-based vehicle routing, but the study also reported limitations related to quantum-hardware scalability, qubit connectivity, and noise. Frequent route updates could also cause route flickering, where vehicles receive different route recommendations before completing their current route. In addition, the formulation mainly considered overlap among the managed vehicle routes and did not directly include pedestrian or general event-attendee traffic [3].

4.3 Computational Considerations

The QUBO formulation involves pairwise interactions between route-selection variables, giving an approximate construction complexity of $O(n^2)$ for a dense representation. The study also reported response time associated with its cloud-based quantum service. These computational and scalability issues are relevant when applying the approach to larger event-based traffic problems [3].

5. PROPOSED SYSTEM

5.1 Quantum Optimization

The study developed a pre-event traffic planning framework for managing the anticipated vehicle movements related to a large public event. An IPL match at the M.A. Chidambaram Stadium, Chennai, was used as the experimental scenario. The framework is intended to help traffic authorities make traffic management decisions prior to the event by identifying appropriate combinations of route diversions, signal timing adjustments, and temporary traffic restrictions[10]. Therefore, the optimization is conducted as a planning exercise rather than as a real-time traffic control mechanism. A predefined event scenario, a modeled road network, and a set of operational constraints are utilized to evaluate various traffic management strategies. This approach allows different traffic management options to be compared before the event, helping authorities understand their possible effects on traffic flow. The results can then support the selection of practical strategies that reduce congestion and improve traffic movement during the event.

Quantum Traffic Optimizer - Overall Architecture

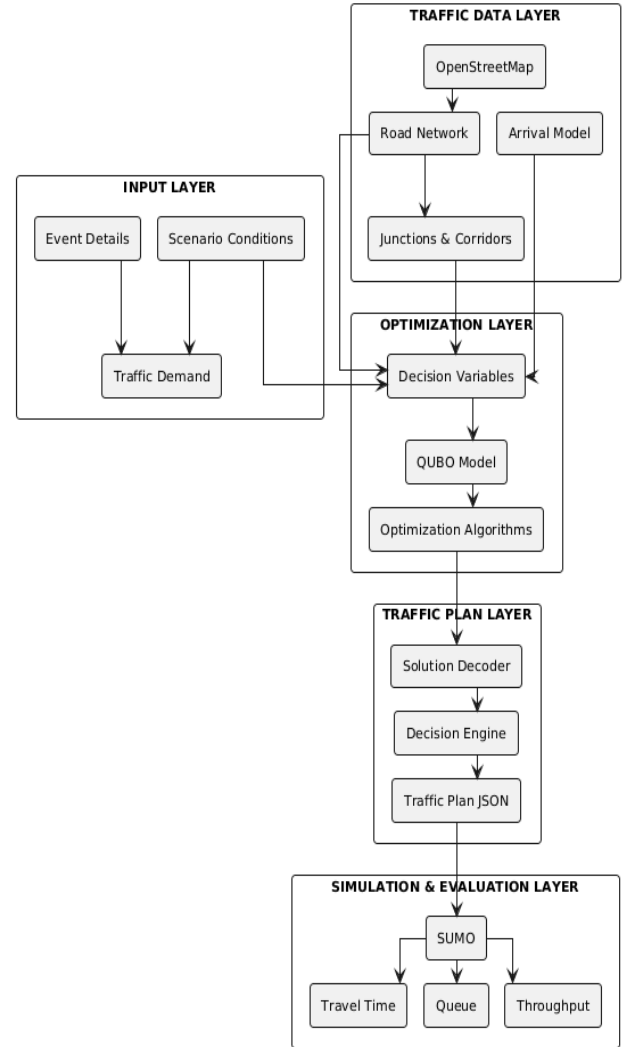


Figure 1: Overall System Architecture

In Figure 1 of Overall System Architecture the Experimental Scenario was established using total traffic-demand boundary of 9515 vehicles for the chosen event. Such criterion was selected as an experimental input on the scenario level to define a scale of a traffic condition imposed by the optimization framework. It must be emphasized that the selected value was not presented as a field-measured traffic count or a number of vehicles observed at the stadium. Instead, it establishes a general demand considered for the scenario, from which a particular part was explicitly defined for the selected network entry points, for the optimization experiment. This approach enables the study to evaluate the proposed optimization methodology under a consistent and controlled traffic-demand

condition while distinguishing the scenario input from empirical traffic data.

The experimental traffic demand was first determined on the level of the overall event scenario. Next, the demand was distributed to the selected entry points of the road network. In particular, 5 entry points were selected for explicit modeling in the event-area network, namely, J1-J5. Instead of using the number of vehicles that was found in the field measurements for these entry points, a certain traffic demand (in terms of number of vehicles) was set to each entry point J1-J5 in order to describe different levels of traffic loading at different approaches to the event area. In particular, the following demands were set: J1 – 1200, J2 – 1447, J3 – 600, J4 – 1100, J5 – 450. By setting different demands to different entry points J1-J5, the optimization model is no more using the same level of traffic loading for all approaches to the event area. Therefore, the demand that was explicitly set to each of the entry points J1-J5 is as follows:

$$D_{\text{modelled}} = D_{j1} + D_{j2} + D_{j3} + D_{j4} + D_{j5} \quad - \text{Eq(1)}$$

$$D_{\text{modelled}} = 1200 + 1447 + 600 + 1100 + 450 \quad - \text{Eq(2)}$$

$$D_{\text{modelled}} = 4797 \text{ vehicles} \quad - \text{Eq(3)}$$

The total scenario-level demand of 9515 was kept intact while 4797 was particularly allocated to the five modeled entry points. This distinction was made for the experimental demand not to be confused with either a traffic count that had been measured or observational data on some roads. The demand for the other scenarios was considered in larger experimental representation instead of being artificially distributed among the five entry points.

As the vehicles that are linked to the large public event would arrive with some timing, rather than all at once, the model was applied with a component accounting for this temporal arrival. A part of the vehicles linked with the event that comes during every interval was represented by a previously defined scenario-based profile. Let p_t represent the share of vehicles arriving during the t -th interval. The summation of the shares for the entire planning period is assumed to be limited by :

$$\sum_t p_t = 1 \quad - \text{Eq(4)}$$

The demand assigned to each time interval is consequently represented as

$$D_t = D_{\text{total}} p_t \quad - \text{Eq(5)}$$

where D_t represents the demand during time period t , D_{total} signifies the demand utilized in the temporal model and p_t expresses the ratio of that demand expressed in the corresponding period. The interpretation appears to be more similar to the behavioral model of such phenomena. The temporal profile has not been considered as a reliable source of empirical data because there are no evidence collected in the field about arrivals in the current experimental regime. The routing surrounding the event area was obtained from OpenStreetMap using OSMnx library in Python. In the practical experiment, the radius of the network was established to be approximately 1200 m away from M. A. Chidambaram Stadium. The acquired routing was illustrated as a graph with nodes being the intersections and edges being the representation of road segments linking the points. The graph in question consisted of about 750 points of intersection and 1985 road segments. By utilizing this graph representation, necessary information on the connections, routes and traffic flows in the event region was obtained.

In this optimization scenario, four key roads were selected as they were associated with the flow of traffic in the area of the event. The roads in question are Anna Salai, Triplicane High Road, Wallajah Road and Kamarajar Salai with number of intersections being 22, 70, 25 and 14 respectively. The roads in question were looked at as potential locations for route traffic interventions and the number of intersections was looked at as potential locations for modifying traffic signal timing. Therefore, the selection of the roads was based on the relevant configuration of the road network that were looked at in the event under quantification.

Traffic capacity was considered as a variable in the modeling. For the geographical and connectivity perspective of the network, OpenStreetMap was used, while the capacity figures utilized in the modelling process were treated merely as modelling variables rather than actual data from the field. The relationship between demand and capacity was modelled through the demand to capacity ratio

$$R = D / C. \quad - \text{Eq(6)}$$

Where D is the traffic demand and C is the associated capacity. If $R > 1$ then this means that the demand is higher than the capacity and if $R < 1$ then this implies that the capacity is more than the demand. The network average capacity and the corridor operational capacity were considered to be two different concepts because otherwise, it could have been misleading to consider the network average as the corridor capacity.

The traffic-management problem was formulated as a binary optimization problem in which every possible intervention was represented by a binary decision variable [14]. For each intervention i , $x_i \in \{0, 1\}$

$x_i = 1$ denotes selection of intervention i while $x_i = 0$ denotes non-selection of intervention i . In total, 18 decision variables were used. The first four decision variables, x_1 to x_4 , corresponded to route diversion decisions with respect to the four chosen corridors. The next ten decision variables, x_5 to x_{14} , corresponded to signal extension decisions for intersections J1 to J10 respectively [11][15]. The last four decision variables, x_{15} to x_{18} , corresponded to temporary restriction decisions with respect to the four chosen corridors [17]. Thus the decision variable vector could be written as

$$x = [x_1, x_2, \dots, x_{18}]^T \quad \text{-Eq(7)}$$

Operational constraints were introduced to limit the number of interventions that could be selected simultaneously. A maximum of two route diversions was permitted, expressed as

$$x_1 + x_2 + x_3 + x_4 \leq 2 \quad \text{-Eq(8)}$$

Similarly, the number of signal extensions was restricted to a maximum of five:

$$x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} \leq 5$$

-Eq(9)

The number of temporary restrictions was limited to one:

$$x_{15} + x_{16} + x_{17} + x_{18} \leq 1 \quad \text{-Eq(10)}$$

The above constraints have been applied to ensure that the final traffic management strategy remains operationally feasible and does not select too many interventions at once for the optimization procedure.

The traffic management formulation took into consideration different dimensions of the resulting traffic condition such as travel time, length of queue, capacity, congestion-based variables, safety variables, and cost of implementation. The initial objective for traffic performance conceptually was represented as

$$F(x) = w_t T(x) + w_q Q(x) - w_r R(x) + P(x) \quad \text{-Eq(11)}$$

Quantum Traffic Optimizer - Optimization Architecture

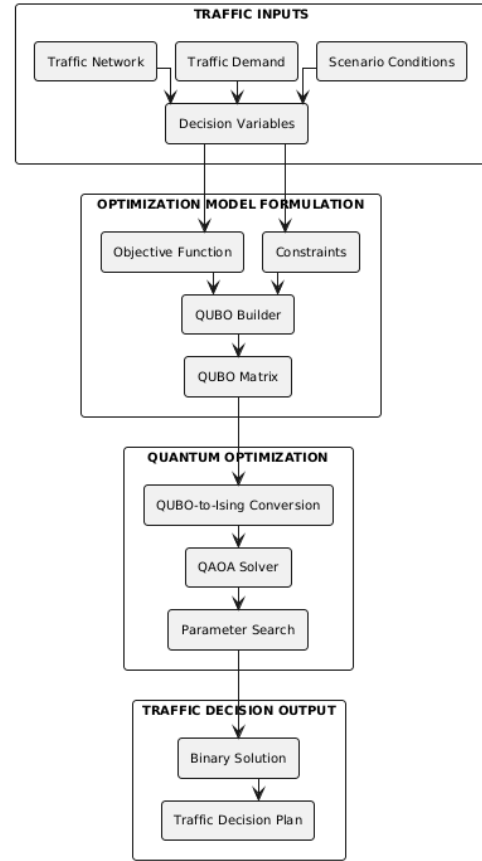


Figure 2: Quantum Optimization Architecture

In this equation, $T(x)$ represents the travel time, $Q(x)$ represents the costs resulting from queuing, $R(x)$ is the benefit from selecting the intervention, while w_t , w_q , and w_r are the weighting factors for each, respectively. The $P(x)$ represents the penalty for constraint violations.

The initial performance objective for traffic and the QUBO energy were two separate entities in this study's methodology. The traffic performance formula is based on the actual traffic management problem in operation, while the QUBO was the mathematical optimization model fed into the optimization phase. The QUBO that was formulated included the individual decision contribution, pairwise decision interactions, cost of implementation, scenario rewards, and penalties.

Converting the constrained binary optimization problem into the unconstrained quadratic problem was achieved using the approach that involved including the operational constraints within the objective

function using penalties. The QUBO formulated took the form

$$E(x) = \sum_i a_i x_i + \sum_{i < j} b_{ij} x_i x_j + P(x) \quad \text{-Eq(12)}$$

where a_i is the linear effect of the decision variable x_i , b_{ij} is the interaction term of the decision variables x_i and x_j , and $P(x)$ is the penalty effect for violating the constraints.

In the case of the applied traffic optimization problem, the QUBO model involved 18 binary variables that could be defined using a matrix of size 18 x 18. The formulation included 18 linear terms and 153 quadratic interaction terms with the use of penalty terms for three operational constraints. The QUBO matrix could be written as

$$E(x) = x^T Q x \quad \text{-Eq(13)}$$

where x is the binary decision vector and Q is the QUBO coefficient matrix. Diagonal entries in Q denote the contribution of the decision variables individually, whereas off-diagonal entries signify interactions between two decision variables. Quadratic interactions permit the inclusion of the information in the model that the effect of one traffic management strategy depends upon other traffic management strategies being included as well.

For the purpose of using the quantum optimization method, all the binary variables were converted into spin variables with the transformation

$$x_i = (1 - s_i) / 2 \quad \text{-Eq(14)[20]}$$

with

$$s_i \in \{-1, +1\} \quad \text{-Eq(15)}$$

The resulting formulation was expressed using an Ising Hamiltonian of the form

$$H_c = \sum_i h_i Z_i + \sum_{i < j} J_{ij} Z_i Z_j \quad \text{-Eq(16)}$$

Here, h_i is the weight associated with the individual variable, J_{ij} is the weight of interaction between the two variables, and Z_i is the Pauli-Z operator for the respective qubit. This conversion changes the binary optimization problem to the Hamiltonian format used in the QAOA algorithm [19]. The conversion of the optimization problem to the Hamiltonian format is a purely mathematical procedure.

Quantum optimization step was designed using Quantum Approximate Optimization Algorithm (QAOA) [16]. As there are 18 binary decision variables in the problem, 18 qubits were utilized in designing the quantum algorithm. An equal superposition state of the computational basis states was considered as the initial quantum state. An initial quantum state for a single qubit can be expressed as

$$|+\rangle = (|0\rangle + |1\rangle) / \sqrt{2} \quad \text{-Eq(17)}$$

The initial state for the complete 18-qubit system is therefore

$$|\psi_0\rangle = |+\rangle^{18} \quad \text{-Eq(18)}$$

The 18 binary variables generate a total search space of

$$2^{18} = 262,144 \quad \text{-Eq(19)}$$

possible binary configurations. The QAOA approach creates a process whereby these possible states can be assigned certain amplitudes that will allow the optimization process to lead to states with low cost Hamiltonian values.

Although exhaustive classical enumeration remains possible for the 262,144 configurations in our present 18-variable configuration, QAOA provides a quantum-based optimization that may be investigated on problems extending beyond this set. Its role in the present study is therefore to assess the applicability of quantum optimization to the formulated traffic-management problem, while the current size of the problem allows for direct comparison to classical enumeration and optimization.

The Ising Hamiltonian obtained from the QUBO was used as the cost Hamiltonian of the QAOA formulation. The corresponding cost unitary operator was defined as

$$U_c(\gamma) = e^{-i\gamma H_c} \quad \text{-Eq(20)}$$

where γ represents the variational parameter associated with the cost layer. A mixing Hamiltonian was defined using the Pauli-X operators as

$$H_M = \sum_i X_i \quad \text{-Eq(21)}$$

The corresponding mixing operator was defined as

$$U_M(\beta) = e^{-i\beta H_M} \quad \text{-Eq(22)}$$

where β represents the variational parameter associated with the mixing layer. For a single QAOA layer, the resulting state can therefore be expressed as [21]

$$|\psi(\gamma, \beta)\rangle = U_M(\beta)U_C(\gamma)|\psi_0\rangle \quad \text{-Eq(23)}$$

For p QAOA layers, the state is generalized as

$$|\psi(\vec{\gamma}, \vec{\beta})\rangle = U_M(\beta_p)U_C(\gamma_p) \cdots U_M(\beta_2)U_C(\gamma_2)U_M(\beta_1)U_C(\gamma_1)|\psi_0\rangle \quad \text{-Eq(24)}$$

The variables γ and β set the transformation performed by the cost and mixing layers. The fitness of a particular setting of the parameters can be assessed based on the expectation of the cost Hamiltonian:

$$\langle H_C \rangle = \langle \psi(\vec{\gamma}, \vec{\beta}) | H_C | \psi(\vec{\gamma}, \vec{\beta}) \rangle \quad \text{-Eq(25)}$$

This process tries to find settings where the expected cost is minimized and the likelihood of achieving low-energy binary solutions is maximized.

In the current approach, the QAOA solution was simulated classically via a grid search on the variational parameters. Therefore, the results obtained for the QAOA approach are simulation results for the QAOA formulation and not those of executing on physical quantum hardware. The current implementation enabled the formulated QAOA to be run with the stated traffic setup.

After the optimal state had been obtained through the optimization process, it was then decoded to determine the appropriate traffic management decisions. If a decision variable $x_i = 1$ for a route-diversion variable, it meant that the respective route had been chosen to be diverted. If a decision variable $x_i = 1$ for a signal-extension variable, it meant that the respective intersection had been chosen to be modified, and if a decision variable $x_i = 1$ for a temporary-restriction variable, it meant that the respective route had been chosen to be temporarily restricted.

The traffic model evaluated the resulting configuration using travel time, queue length, throughput, congestion-related measures, and implementation penalties [18]. Throughput represented the number of vehicles processed by the explicitly modeled network and was therefore distinct from the initial scenario demand of 9,515 vehicles. The implementation penalties were assigned as weights of 8, 3, and 12 for route diversion, signal extension, and temporary

restriction, respectively, with the total penalty determined by the selected decision variables.

The QUBO energy was not considered to be a direct measure of the performance of the traffic. The QUBO energy is the mathematical energy that is optimized in the process. On the other hand, traffic time, traffic queues, and all the other parameters are the result of traffic model calculations, after the chosen traffic management setting is analyzed using the traffic model. Thus, the QUBO objective and traffic performance parameters remained separate measures.

The QAOA-based formulation was benchmarked against two additional formulations of the optimization problem using the same traffic situation and decision problem formulation. The first alternative was based on the application of the greedy technique wherein interventions were performed on an individual basis based on expected gains from each intervention without the joint optimization of all interactions between decisions. The second approach was classical optimization where the same decision variables, objective formulation, and constraints were applied with the use of classical optimization procedures. The QAOA-based technique used the same formulation with QAOA-based parameter search procedure.

For the current experimental case, the greedy algorithm saved about 2.8 minutes of travel time and reduced the queue by 6%, taking about 0.016 seconds of computing time. For the classical optimization method, the travel time was saved by about 3.3 minutes, reducing the queue by 8%, with the computing time being about 2.55 seconds. For the QAOA-based method, the travel time savings was about 3.34 minutes, while the queue was reduced by 8.2%, taking about 18.03 seconds for the computation. This provides an experimental basis of comparing the three approaches within the specified scenario. However, no general quantum supremacy is evident from this experiment, since the QAOA-based approach was executed using classical simulation and the traffic performance differences were noted from a single experimental setting.

The optimized decisions for traffic management were ultimately encoded into a JSON format that contains the diversions, signal timings changes, temporary bans, along with all of their parameters. This format will allow a uniform input channel for passing the output of the optimizer into the next step of traffic simulation. This allows the plan created by the optimizer to be tested separately from the baseline scenario.

This methodology thus includes event-demand modelling using scenarios, temporal arrival modelling, extraction of road networks from OpenStreetMap data, capacity modelling, binary decision modelling, QUBO modelling, Ising Hamiltonian modelling, optimization using QAOA by classical simulation, and traffic performance comparison. This framework offers a systematic means of examining how different route redirections, signal timing changes, and temporary traffic restrictions can be used in a predefined traffic scenario for a major event.

Quantum Traffic Optimizer - Traffic Decision and Simulation Architecture

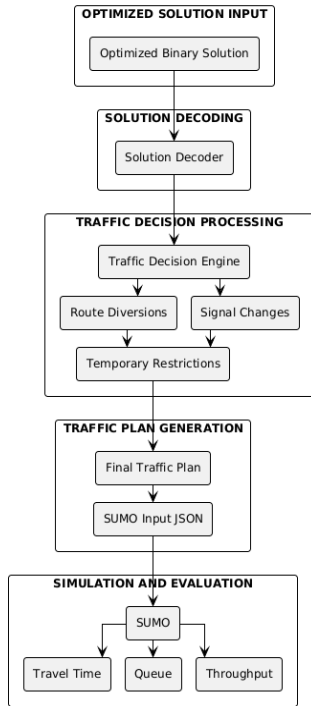


Figure 3: Traffic Decision and SUMO

5.2 Digital Twin

The system is a twin with a closed loop. It simulates traffic, people and trains of the particular district. No real-world data is included. All the data is generated by one simulator SUMO.

There are three layers:

1. Simulation layer (SUMO).

In this layer of road network, lights, cars, people and trains are represented. They all move in steps with step-time 0.5 sec.

2. Controller layer (Python: FastAPI, TraCI, NumPy, AsyncIO):

In this layer the Communication with SUMO via TraCI on a socket connection is done and management of the scenario with generation of crowd burst when train appears. Also calculation of the metrics and generation of JSON snapshot and sending the JSON snapshot via WebSocket with 30 FPS.

3. Visualization layer (Three.js, WebGL):

This layer works on the generation of city by requesting the snapshot and animation of objects .Moving of objects smoothly with LERP to provide 60 FPS and HUD display with KPIs and event phase.

The vehicles follow Intelligent Driver Model (IDM).The acceleration of the vehicle depends on the speed of the vehicle, the speed it wishes to attain, the distance to the vehicle in front, and the relative speed of the vehicle in front.

The sidewalks and plazas have stripes running along their lengths, each having a width of 0.55 m. Pedestrians prefer to walk on stripes overtake slow moving people avoid walking against the main flow of pedestrians and halt at signalized crossings.

The railway track is located at an elevation of 8.0 m on a viaduct supported by pillars after every 40 m. The trains follow the Krauss model and run on a schedule with fixed headways. Trains move slowly while approaching a station halt there for 30–35 s and accelerate again up to 80 km/h. The track is completely elevated. There are no level crossings. The road and rail collisions are avoided because of such construction.

SUMO uses X for East, Y for North, and Z for altitude. In SUMO, a heading of 0° indicates North direction. Increases clockwise. Three.js uses X for East, Y for Up, and Z for South, which is the direction where the viewer looks from. In Three.js, a heading of 0 radians indicates +Z direction. Increases counter-clockwise. The formulas are given as:

$$\text{Position: } (X_{3D}, Y_{3D}, Z_{3D}) = (X_{\text{SUMO}}, Z_{\text{SUMO}}, -Y_{\text{SUMO}})$$

-Eq(26)

$$\text{Rotation: } \theta_{3D} = -\theta_{\text{SUMO}} \pi / 180$$

-Eq(27)

Trains are displayed at an elevation of 8.1 meters. Cars and pedestrians are displayed at an elevation of 0.25 meters.

The Vehicle Motion: Intelligent Driver Model (IDM).
For vehicle a, when followed by a vehicle a-1:

$$dv/dt = a[1 - (v/v_0)^{\delta} - (s^*(v, \Delta v)/s)^2] \quad \text{-Eq(28)}$$

$$s^*(v, \Delta v) = s_0 + vT + v\Delta v/2\sqrt{ab} \quad \text{-Eq(29)}$$

$$s = x_{a-1} - x_a - L_{a-1} \quad \text{-Eq(30)}$$

$$\Delta v = v_a - v_{a-1} \quad \text{Eq(31)}$$

Symbol	Meaning	Value
v_0	Desired speed	13.89m/s (Major roads) 11.11 m/s (Minor roads)
a	Maximum acceleration	2.6 m/s ²
b	Comfortable deceleration	4.5 m/s ²
Δ	Acceleration exponent	4
s_0	Distance at which jam occurs	2.0 m
T	Safe headway time	1.2 s

Table 2. Parameter Settings Used in Vehicle Motion.

Event Day demand is a piecewise non-homogeneous Poisson process that takes place in four phases.

Phase 1 Pre-event build-up: ($0 \leq t < T_1$):
 $\lambda(t) = L_{base} + (L_{surge} - L_{base}) * (t / T_1) \quad \text{-Eq(32)}$

Phase 2 Arrival surge: ($T_1 \leq t < T_2$):
 $\lambda(t) = L_{surge} + L_{stadium_inflow} \quad \text{-Eq(33)}$

Phase 3 Peak match: ($T_2 \leq t < T_3$):
 $\lambda(t) = L_{peak_match} \quad \text{Eq(34)}$

Phase 4 Post-event egress: ($t \geq T_3$):
 $\lambda(t) = L_{base} + L_{egress} * \exp(-(t - T_3) / \tau) \quad \text{-Eq(35)}$

Constants:

$T_1 = 300$ s, $T_2 = 900$ s, $T_3 = 1500$ s. τ represents the depletion time constant.

The inflow to the Central Stadium Road and drop-off area is over 600 devices per hour.

The complexity of computations can be expressed as follows:

N_v - number of vehicles in motion. N_p - number of people on foot. E - number of street segments. J - the number of traffic lights. C - the number of clients connected. The update rate of the simulation is at 2 Hz (with an update interval of 0.5 s). The update rate of rendering is 60 Hz.

Space Complexity can be described by:

1. The usage of the memory on the server:

$O(N_v + N_p \cdot E \times H)$ (where $H = 120$ steps of history). The amount of memory occupied by SUMO = 45 MB. In combination with Python and metric memory usage the amount of memory is around 35 MB.

2. The amount of bandwidth consumed:

The size of the Event Day Snapshot Peak (250 vehicles and 300 people) = 18 KB. In the case of 30 Hz: 18 KB times 30 = 540 KB per second or 4.32 Mbps.

3. Memory consumption on the client's GPUs:

110 MB (considering vertex buffers, assembly buffers, and shaders.). Up-to-date methods of doing things come with some questions regarding their initial speed, scalability, and the application fields

themselves. Firstly, the problem lies in the TraCI communication overheads. Whenever different methods from the TraCI library are called - or `getAngle`, `getSpeed` and `getPosition` functions within TraCI are called, the number of exchanges N_v between the simulation application and the simulation program emerges. Thus, the overhead increases with the number of cars in the simulated system. However, the problem can be resolved through the `traci.vehicle.subscribe()` or `subscribeContext()` methods.

Another restriction that can appear is the overuse of JSON for communicating purposes. The usage of JSON suggests that certain CPU cycles will be used for serializing and parsing. What is more, the speed of the system is also decreased because of the number of actors.

Thus it is necessary to implement binary serialization systems such as Protocol Buffers or FlatBuffers because they allow cutting down on data and, accordingly, communication overhead.

Another problem with the -batched WebGL draw calls occurs within this particular system. The more cars and pedestrians are displayed through the meshes the more draw calls occur. It can reach up to 1500 draw calls per one frame. This is problematic for the end and mobile GPUs in terms of rendering performance. This can be solved by the implementation of the object pooling and GPU instancing technologies. For instance, the use of `THREE.InstancedMesh` allows rendering objects much better when their number is higher than 2000.

The other limitation is that it takes into account paths for pedestrians. The pedestrians have to stick to the paths provided to them and cannot change their location randomly within the large area around them. Consequently, pedestrian movement does not appear natural in the simulation. It can be addressed by applying either Social Force model or NavMesh technology.

Another flaw of the current model is the artificial generation of traffic demand. Since no calibration of any OD matrices takes place and there is not enough real-world ground truth statistics used in the simulation, it can be presumed that the generated traffic will not resemble real traffic at all. Thus, there is no way of validating the traffic statistics generated. However, the problem can be solved by the use of open and calibrated OD data along with validation of simulated traffic statistics via sensors.

Finally, the model is run on the basis of a feeder assumption. The traffic enters the simulation network through certain edges like , , etc. Traffic, that is outside the simulated region, does not appear directly in the simulation; thus, all external traffic interactions and traffic spillbacks are not completely simulated. This issue in the model could be resolved either by expanding the boundaries of the network or by connecting the simulation to a regional traffic simulation model.

6. RESULT

The traffic optimization framework proposed for the event was assessed based on five different scenarios characterized as good weather, heavy raining, movement of VIPs, road construction, and combinations of various situations. The framework was assessed by using three optimization techniques, namely greedy optimization, classical optimization, and QAOA-based optimization.

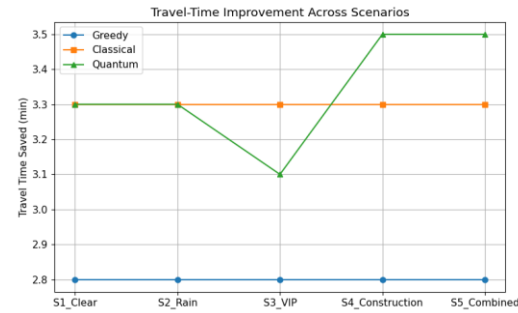


Figure 4: Travel Time Improvement Access Scenarios

The greedy was constant at 2.8 minutes saved for every experiment. Classical was constant at 3.3 minutes for all experiments. QAOA was 3.3, 3.1, 3.5 minutes for the clear/heavy-rain, VIP, construction/combined cases respectively. QAOA was equivalent to classical for clear/heavy-rain, 0.2 minutes worse for VIP, and 0.2 minutes better for construction/combined scenarios.

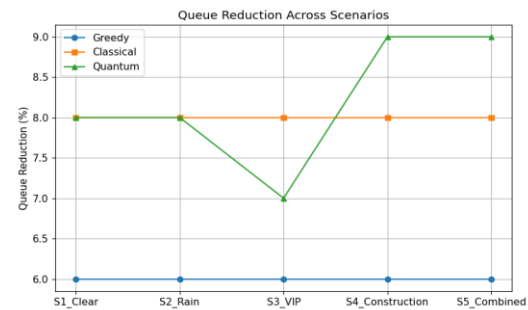


Figure 5 : Queue Reduction Across Scenarios

The second metric considered was reduction of queues. When using the greedy approach, the average number of queues reduced per case was 6%. The classical approach averaged 8% reduction overall. The QAOA based approach achieved 8%, 7%, and 9% reduction under clear and heavy-rain, VIP, and construction and all scenarios, respectively.

When used with the case based approach to manage queues, the QAOA approach achieved 8% reduction compared to the classical approach when managing queues in all conditions and scenarios. QAOA achieved 9% reduction with a case based approach in the combined scenario whereas the classical approach achieved only 8%. It appears as though the two approaches are able to handle the three operational constraints in the same way. The travel-time results for scenario-level cases lend themselves to the same conclusion as the queue results.

When the classical approach is used the travel-time is 3.3 min in every case. When the QAOA based approach is used in every case it is 3.1 min in the VIP case and 3.5 min in every other case. There is a difference in the case based approach to manage queues for the QAOA approach to manage queues. Overall the three approaches used in this work each gained 2.80, 3.30, and 3.34 min in average travel-time savings in all scenarios and gained 6.0, 8.0, and 8.2% in queue reduction for greedy, classical, and QAOA-based approaches, respectively.

Compared to the greedy baseline, QAOA achieved average improvements of 19.286% in travel-time saving and 36.667% in queue reduction, whereas the classical approach achieved 17.857% and 33.333%, respectively. All means are directly reported in the efficiency report. Improvements of QAOA over the classical approach were small. 0.04 min in average travel-time saving and 0.2 percentage points in average queue reduction. So the main conclusion is that the QAOA-based approach reached traffic-performance results very close to the classical approach, rather than algorithmic dominance.

The QUBO objective values were also investigated to assess the behavior of the mathematical optimization formulation. The greedy and classical objective values were constant at five scenarios at approximately -528.28 and -601.68, respectively. On the other hand, the QAOA-based objective changed with the scenario, taking values of -469.25 for the clear scenario, -489.00 for heavy rain, -461.20 for the VIP scenario, -489.50 for construction, and -425.25 for the combined scenario.

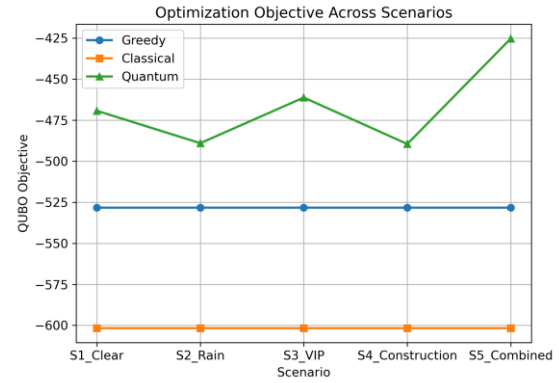


Figure 6 : Optimization Objective Across Scenarios

The average QUBO objective values were -528.28 for greedy, -601.68 for classical and -466.84 for the QAOA-based approach. Since the QUBO objective represents the objective of the mathematical optimization problem, these values have to be considered separately to traffic-performance indicators, as travel-time saving and queue reduction. In the current formulation, the classical method achieved a numerically lower QUBO objective, while the QAOA-based method achieved competitive traffic-performance outcomes. This shows that the optimization-objective value and downstream traffic-performance indicators are related but non-interchangeable evaluation metrics.

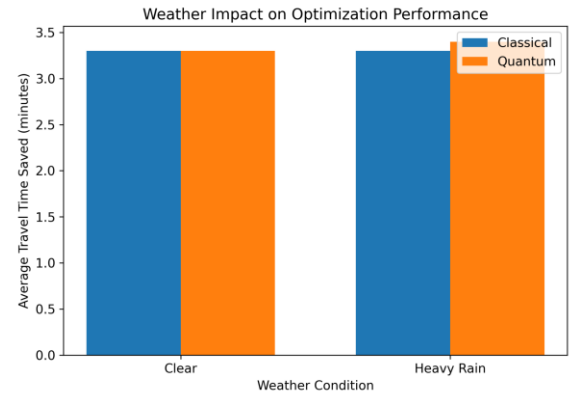


Figure 7 : Weather Impact on Optimization Performance

The impact of weather conditions was analyzed by comparing the clear and heavy-rain scenarios. For the clear setting, the classical and QAOA-based approaches both resulted in 3.3 minutes of travel-time saving. The same was observed for the heavy rain condition, where the classical approach produced 3.3 minutes of travel-time saving, and the QAOA-based

method produced 3.3 minutes as well. Correspondingly, both the approaches reduced the queues by 8% each in both the settings.

Based on the weather-specific analysis, it can be concluded that the QAOA-based approach was able to maintain the same level of traffic-performance as that of the classical approach when the scenario was switched from clear weather to heavy rain. This provides evidence that the optimization formulation was able to remain stable under the environmental change that was encapsulated within the experimental scenarios.

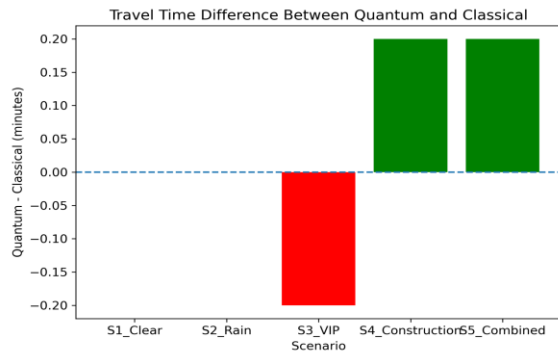


Figure 8: Travel Time Difference Between Quantum and Classical Algorithm

A direct comparison between the QAOA-based and the classical optimization further highlighted these scenario-dependent results. The differences in travel time was zero for the clear and heavy-rain scenarios, 0.2 min for the VIP scenario, and +0.2 min for the construction and combined scenarios. Therefore, the QAOA-based optimization was neither better nor worse than the classical optimization but instead depended on the constraints being applied in the scenarios.

This was especially true in the combined scenario where the maximum number of simultaneous constraints was present, and in which the QAOA based gave a travel-time saving of 3.5 min and queue reduction of 9% compared with 3.3 min and 8% for classical optimization, corresponding to QAOA based gave the highest values of trafficperformance for QAOA in this study.

The experimentally observed outcomes suggest that the QAOA-based formulation successfully obtained performance equivalent to classical optimization across the five pre-event traffic scenarios. The QAOA-based optimization resulted in an average of 3.34 min saving in travel time (compared to 3.30 min for the

classical optimization) and an average of 8.2% reduction in queue length (compared with 8.0%). Based on the classical optimization method, QAOA achieved a 19.286% improvement in terms of the average travel-time saving, and a 36.667% improvement in terms of the average queue reduction over the greedy method. At the level of individual scenarios, the QAOA based achieved the same outputs as classical method in the clear and heavy-rain scenarios, led to slightly lower traffic management outputs in the VIP scenario, but outperformed classical optimization in the combination of the two scenarios (QAOA gave 3.5 min travel-time saving and 9% queue reduction whereas classical optimization gave 3.3 min and 8%, respectively).

Analysis of the computer simulation demonstrated a different trade-off than in the scenario analysis. The QAOA implementation took an average of 18.03 seconds, versus 2.55 seconds for the classical optimization, and had a lower overall efficiency score than the classical implementation. That result makes sense given that the current QAOA implementation was tested on a classical computer simulation rather than a real quantum computer.

The Digital Twin was built for the MA Chidambaram Stadium (Chepauk) region using SUMO as the microscopic traffic engine. The network covers the main roads, junctions, traffic signals, pedestrian paths, and the public-transport elements around the stadium. Each vehicle in SUMO moves on its own, so queues and congestion form from individual driving behavior and not from fixed averages. Figure 9 shows the SUMO network used in the experiments. The stadium is marked in the middle of the study area, with the major roads forming a loop around it. This layout makes the region a good test case, because event traffic has to pass through a small number of junctions.

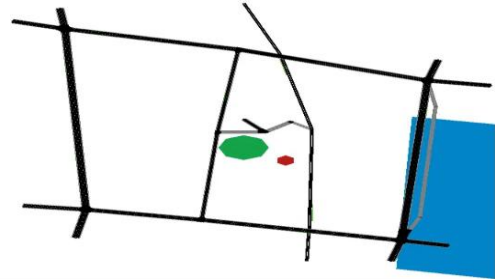


Figure 9: SUMO-based Chepauk Digital Twin network.

Two scenarios were prepared, a Normal Day and an Event Day. The Event Day scenario represents the

heavier traffic seen when a match is on. The demand can be set from the interface for each simulation hour. For the Event Day run, the inputs were 5,540 cars, 45 buses, 1,550 two-wheelers, 2,400 pedestrians, and 80 local trains. These values scale the existing SUMO flow definitions, while the route geometry stays the same. This keeps every run comparable. All results in this section come from the Event Day scenario.

The image shows a 'SCENARIO INPUT' panel with a 'Configure traffic' tab. Under the 'Scenario' section, 'Event day' is selected. Below, it states 'Scenario follows the selected SUMO demand profile.' A table 'Demand per simulation hour' lists vehicle types and counts: Cars (5540), Buses (45), Two wheelers (1550), Pedestrians (2400), and Local trains (80). A note at the bottom says 'Values scale the existing SUMO flow definitions; route geometry stays unchanged.'

Demand per simulation hour		VEH / PERSONS
Cars	5540	45
Buses		
Two wheelers	1550	2400
Pedestrians		
Local trains	80	

Figure 10: Scenario input panel showing Event Day demand per simulation hour.

SUMO produces the traffic state at every simulation step, and TraCI reads this state and passes it to a Python backend. The state includes vehicle positions, traffic-signal conditions, pedestrian activity, and congestion information. The backend then streams it over WebSocket to a Three.js interface, which redraws the 3D scene each time a new frame arrives. During testing, the WebSocket delivered both the road geometry and live simulation frames. The geometry is sent first so the browser can build the road network, and the live frames follow to update the moving elements. One recorded frame, taken at about 437.5 seconds of simulation time, held 260 vehicles, 141 pedestrians, 7 traffic lights, and 60 congestion edges. This confirms that the interface is driven by the running simulation and not by a prepared animation. The numbers also change from frame to frame as traffic builds up.

Figure 11 shows the complete dashboard. The classical baseline and the optimized simulation run side by side, with synchronized cameras. The cameras move together, so the viewer looks at the same location in both views and can see where the two states differ. At the clock reading of 04:26.0, the network held 236 vehicles and 86 pedestrians. The vehicle mix was 174 cars, 3 buses, 57 two-wheelers, and 2 local trains. The average speed was 13.2 km/h, the queue length was 617.5 m, and 17 roads were congested. Road colors follow four congestion levels, from low to

severe, so crowded sections are easy to tell apart from free-flowing ones. The congestion data comes from SUMO itself and is not drawn by hand. The side panel also shows the live metrics, and the bottom panel shows the optimization status and the actions applied.

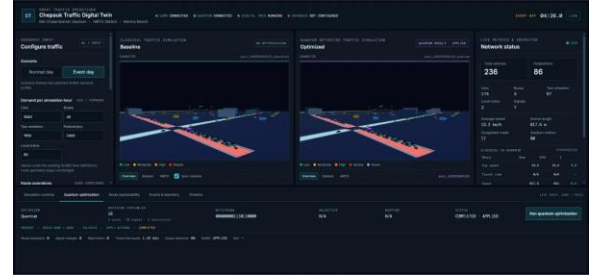


Figure 11: Digital Twin dashboard with the baseline and quantum-optimized simulations, live metrics, and optimization status

The Digital Twin does more than display traffic. It can also act on the running simulation. The optimization problem used 18 decision variables: 4 for routes, 10 for signals, and 4 for restrictions. The solver returned the bitstring 000000001110110000, and the dashboard reported the run as completed and applied.

In the live integration experiment, 23 traffic-control commands were generated, and every target identifier matched an element in the active SUMO network. After the commands were applied, route-related values changed on 18 mapped edges. Signal timing also changed. The next-switch time moved from 36 to 42 seconds at J_CENTRAL_STAD, from 36 to 46 seconds at J_NE, and from 36 to 43 seconds at J_SW.

The optimization model uses logical labels such as Anna Salai and J1, which SUMO does not understand. A mapping layer converts these labels into real SUMO identifiers before any command is issued. All four optimization corridors and the J1–J10 signal identifiers were checked against the Event Day network. This keeps the optimization model separate from the simulation itself.

Figure 12 shows the full sequence. The baseline traffic is observed, optimization is triggered, and the result is turned into traffic actions. These actions become TraCI commands, SUMO updates its state, and the new state is shown in the interface.

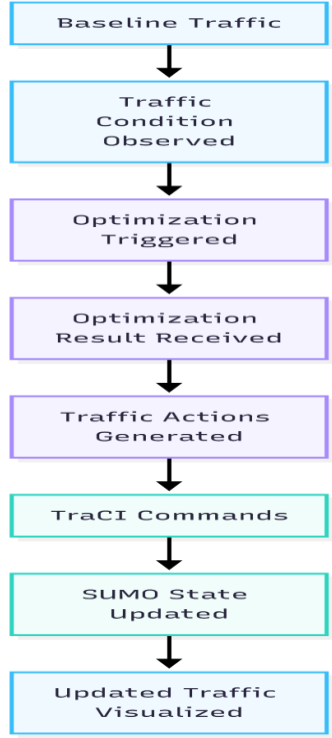


Figure 12: Operational sequence from baseline traffic to the updated visualization.

The side-by-side comparison in Figure 11 shows the effect of the applied result. The average speed stayed at 13.2 km/h in both views, while the queue length fell from 617.5 m to 611 m, a drop of 6.5 m. The dashboard also reported a travel-time saving of 1.25 minutes. These gains are small in this snapshot, which was taken early in the Event Day run. At that point, the network had not yet reached its heaviest load. The main finding here is that the full loop works. A decision made by the optimizer reaches the simulation and produces visible changes. Larger gains can be studied in longer runs and at peak event demand.

The Unicorn-based server started successfully. Its endpoints for geometry, optimization triggering, optimization-run information, and the latest optimization plan all returned successful responses. This shows that the backend is ready to be used by the interface and by the optimization workflow. The browser opened the WebSocket connection, rendered the 3D road network, and received the quantum optimization information through the same state stream. Because everything travels over one stream, the interface always shows the same state as the simulation.

The results confirm three working stages. SUMO generates the traffic behavior, TraCI reads and changes the running simulation, and WebSocket with Three.js turns that state into a live visual interface. Commands from the optimization workflow reached SUMO and changed both route values and signal timings. The identifier mapping also worked on the Event Day network, so abstract optimization labels were applied to real SUMO elements. This makes the Digital Twin a usable platform for testing traffic-management decisions around the Chepauk stadium during events. It gives a safe place to try a decision and see its effect before it is used on real roads.

7. DISCUSSION

7.1 Overall Findings

The framework was tested on an IPL event around the M. A. Chidambaram Stadium, analyzing route diversions, signal extensions and temporary traffic restrictions. QAOA yielded a feasible traffic-management plan close to classical optimization.

7.2 Strengths of the Proposed Framework

The framework brings different traffic-management decisions into one optimization problem. Route diversions, signal extensions, and temporary restrictions can therefore be considered together. The QUBO also includes operational constraints through penalty terms, which helps avoid invalid combinations of decisions. The selected decisions can then be exported for further traffic-simulation evaluation.

7.3 Limitations

In this work, we analyze the pre-event traffic planning based on a target event scenario and several fixed traffic-management measures such as route diversion, signal extension, and road restriction. The assessment was carried out in a simulated quantum environment using synthetic demand, which makes it possible to conduct the effectiveness of our proposed framework. In the future, we can expand the scope by simulating multiple event scenarios, including larger and more varied road networks, live-traffic conditions, and physical quantum hardware.

7.4 Practical Implications

The framework can be used for pre-event traffic planning by allowing different combinations of route diversions, signal adjustments, and temporary restrictions to be evaluated before an event. The selected decisions can also be exported for further testing through traffic simulation. This can help compare alternative traffic-management plans before they are considered for real-world use .

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